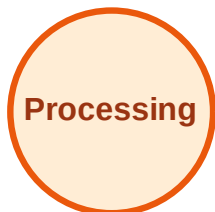


BFS Traversal

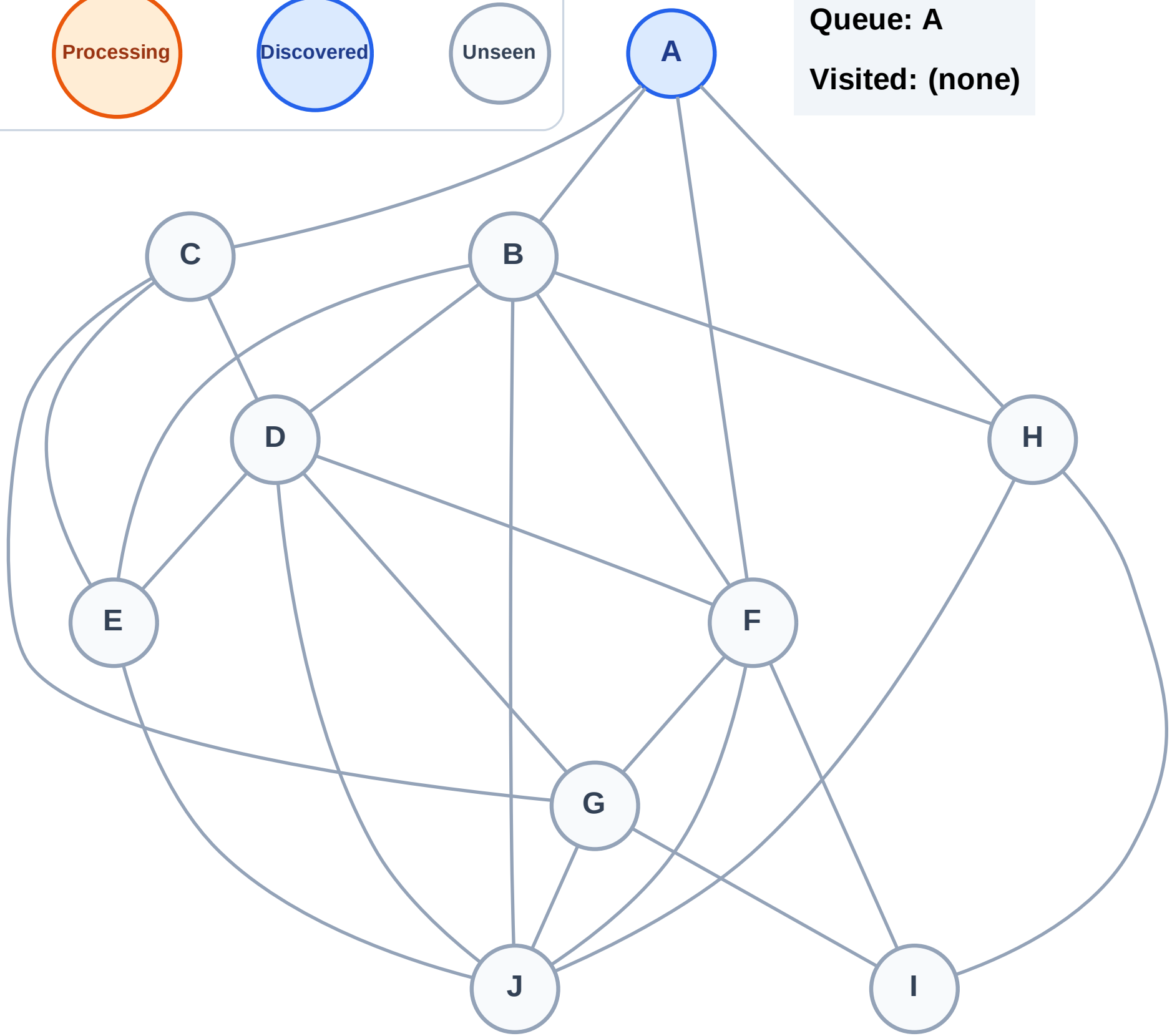
Step 1: Start at A

Legend



Queue: A

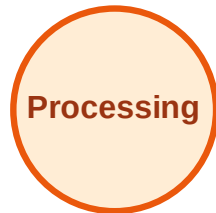
Visited: (none)



BFS Traversal

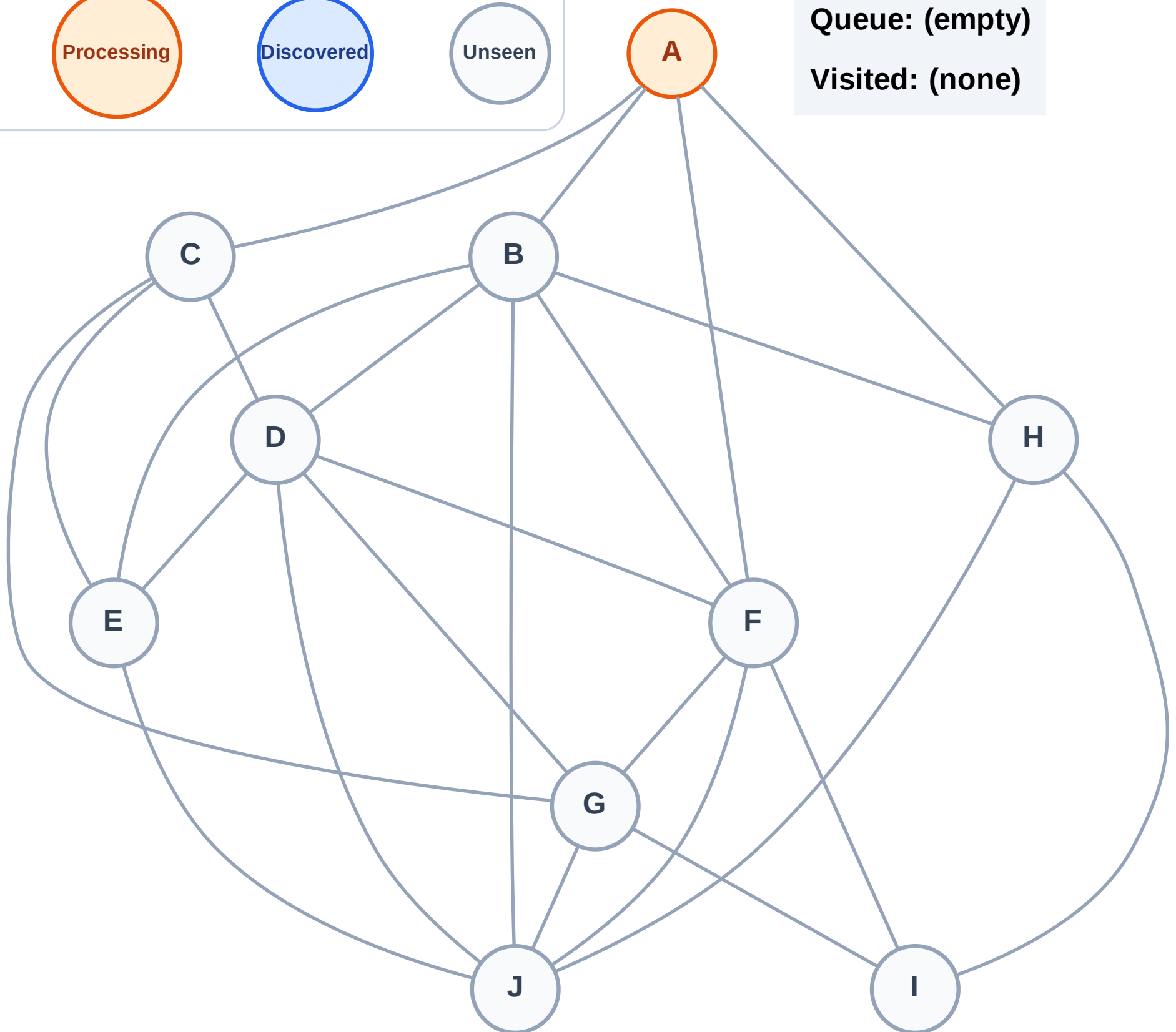
Step 2: Process node A

Legend



Queue: (empty)

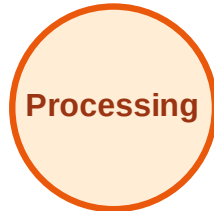
Visited: (none)



BFS Traversal

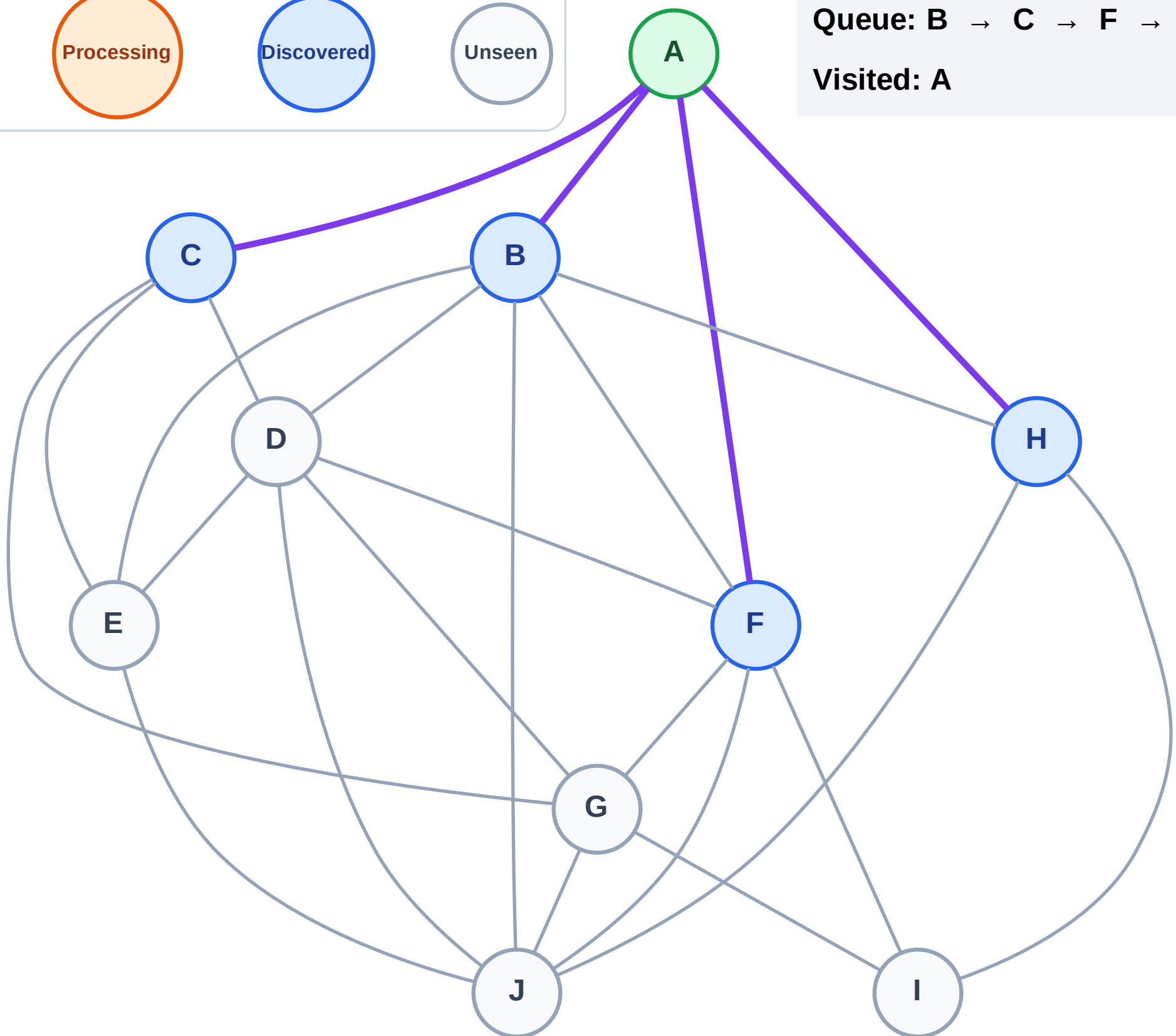
Step 3: Discover B, C, F, H from A

Legend



Queue: B → C → F → H

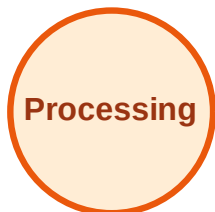
Visited: A



BFS Traversal

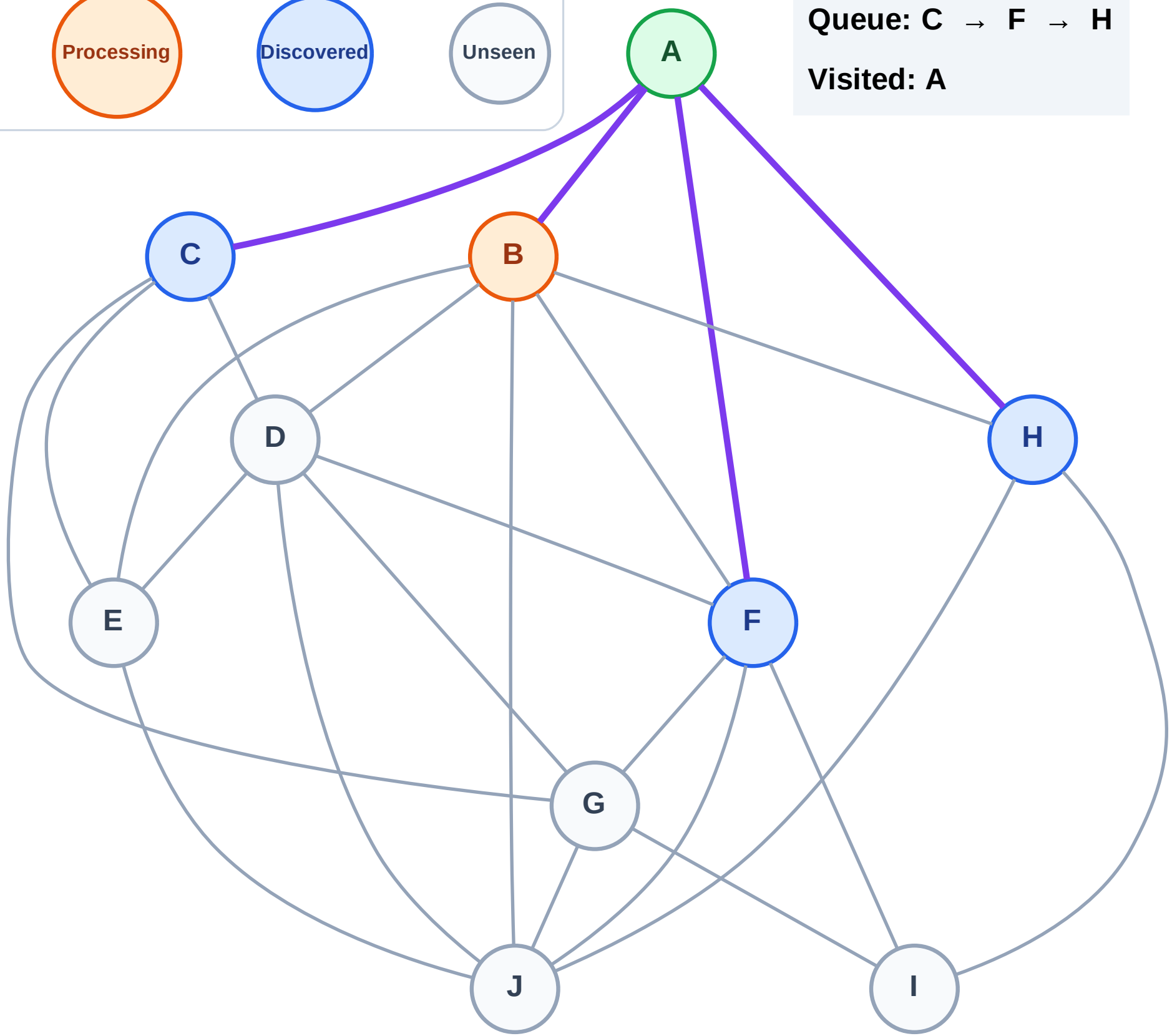
Step 4: Process node B

Legend



Queue: C → F → H

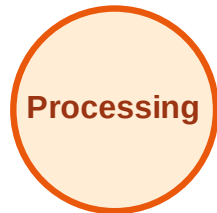
Visited: A



BFS Traversal

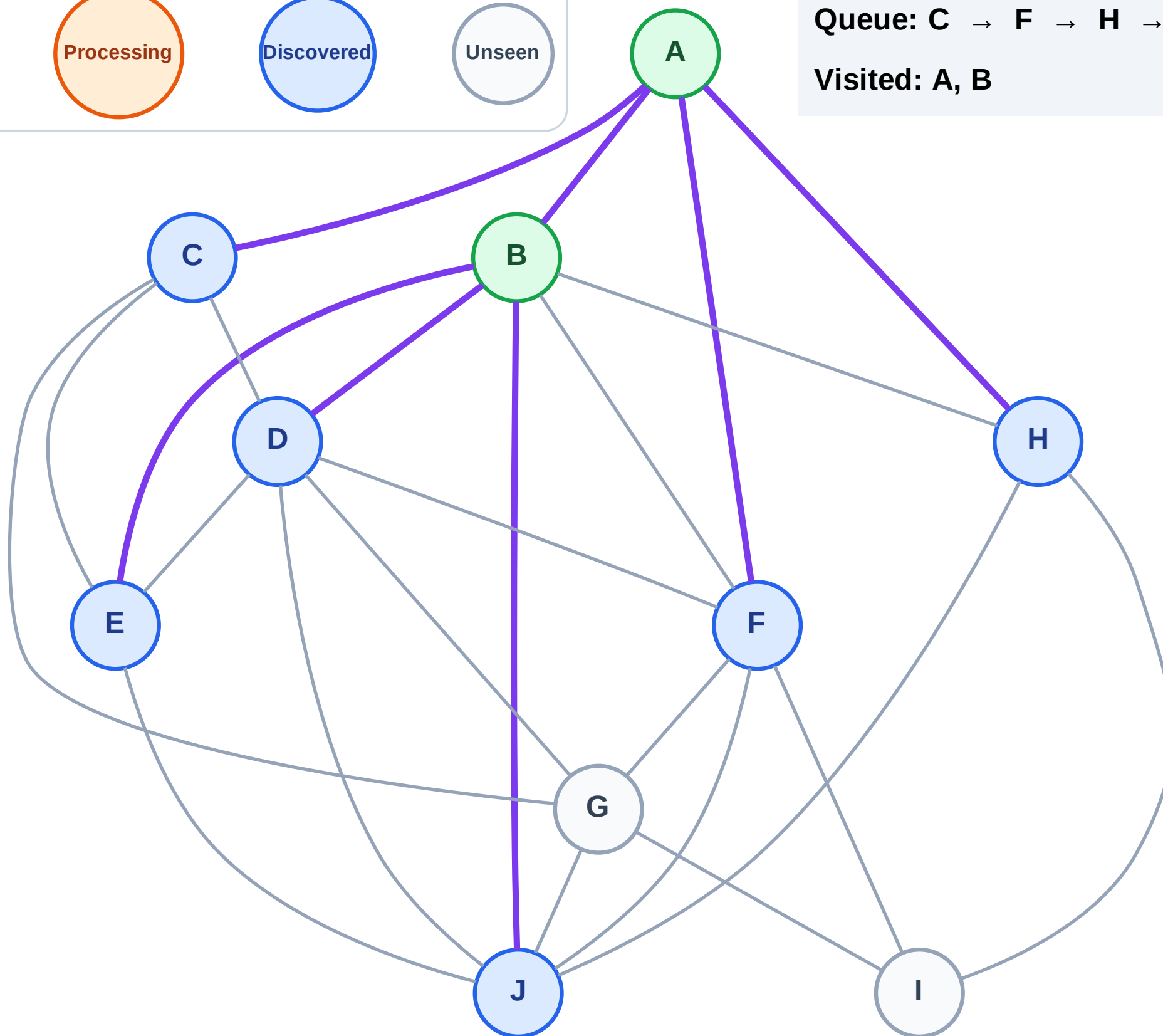
Step 5: Discover D, E, J from B

Legend



Queue: C → F → H → D → E → J

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

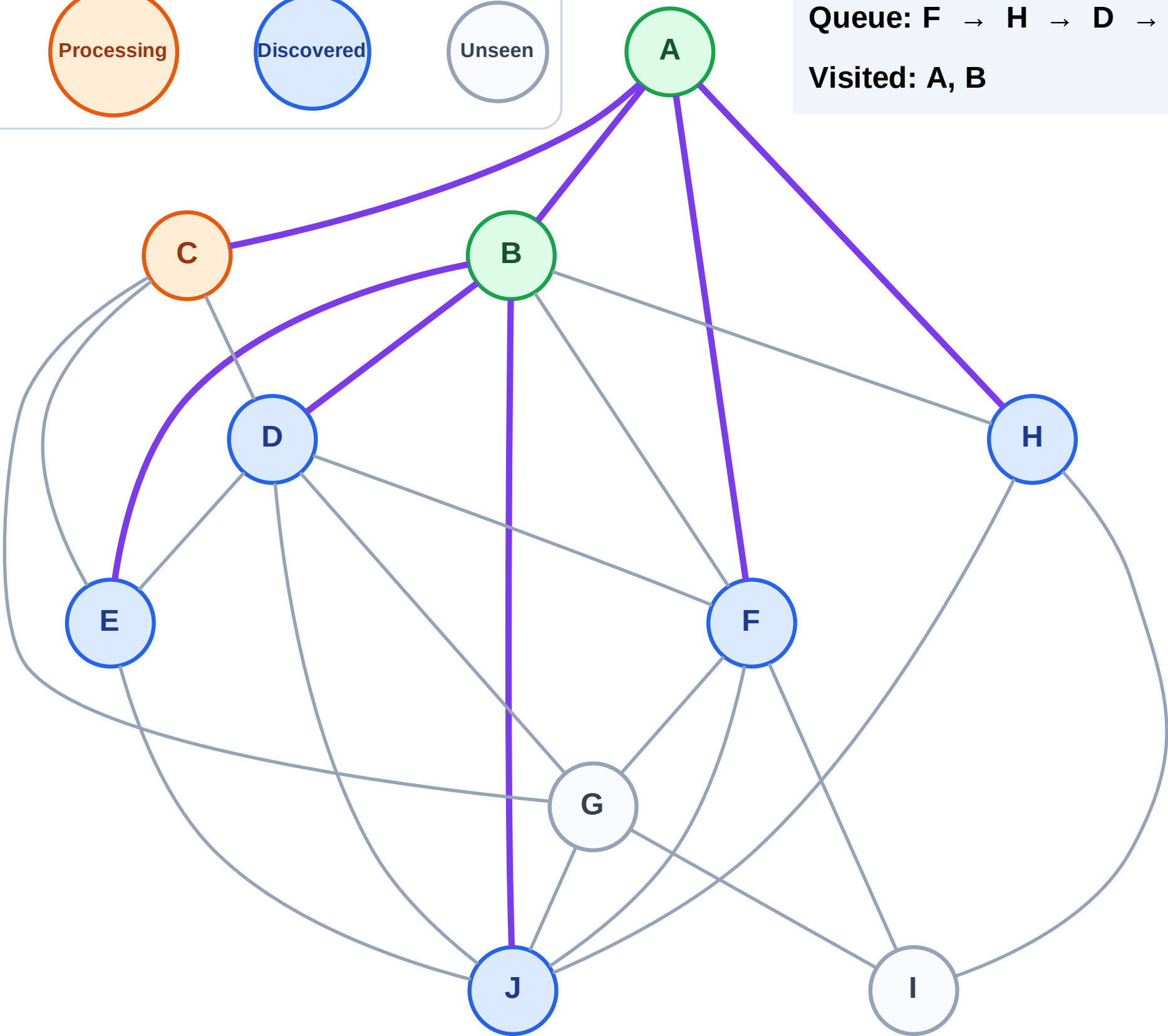
Processing

Discovered

Unseen

Queue: F → H → D → E → J

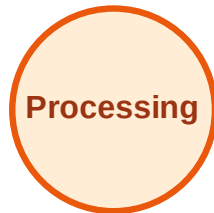
Visited: A, B



BFS Traversal

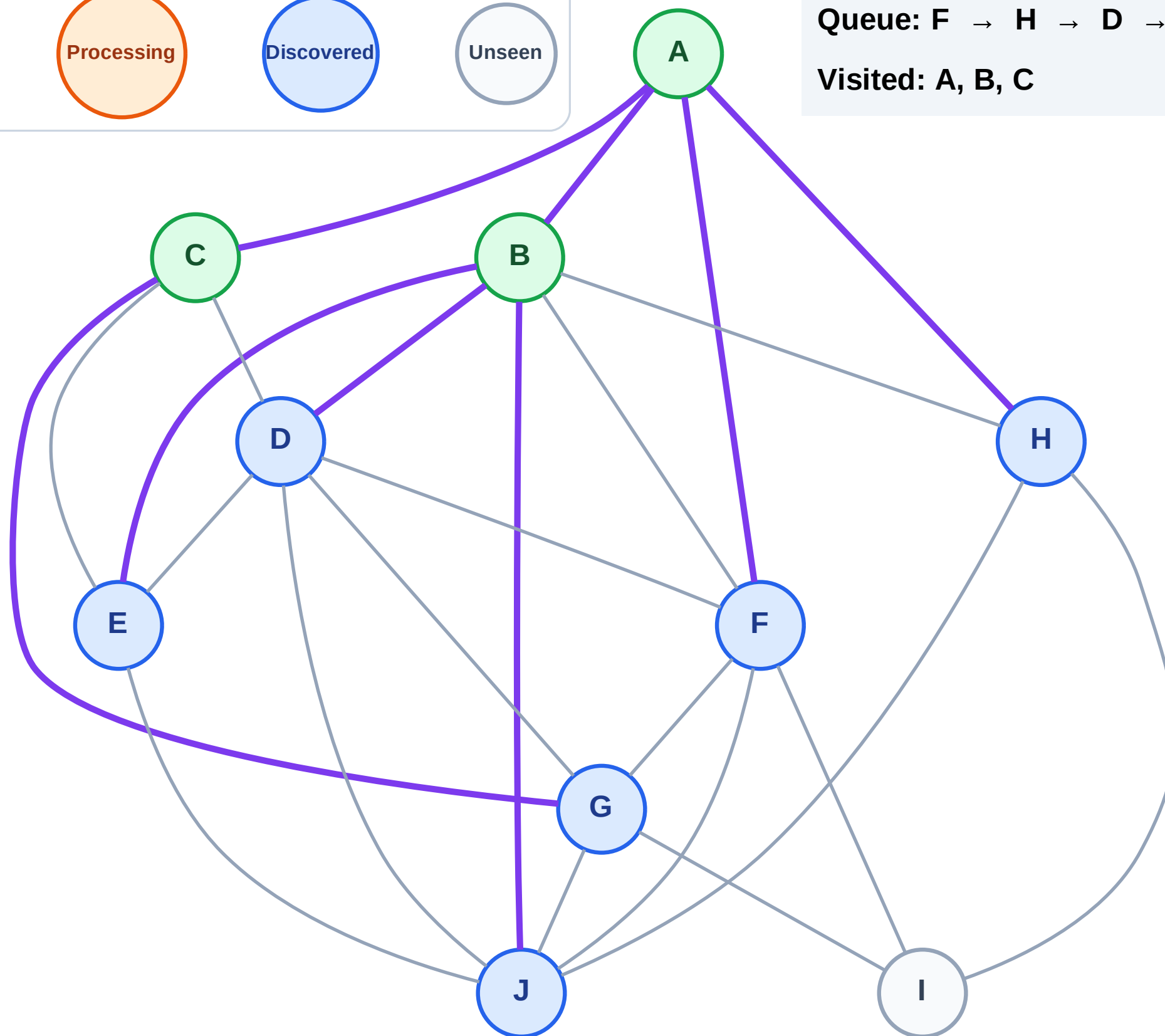
Step 7: Discover G from C

Legend



Queue: F → H → D → E → J → G

Visited: A, B, C



BFS Traversal

Step 8: Process node F

Legend

Complete

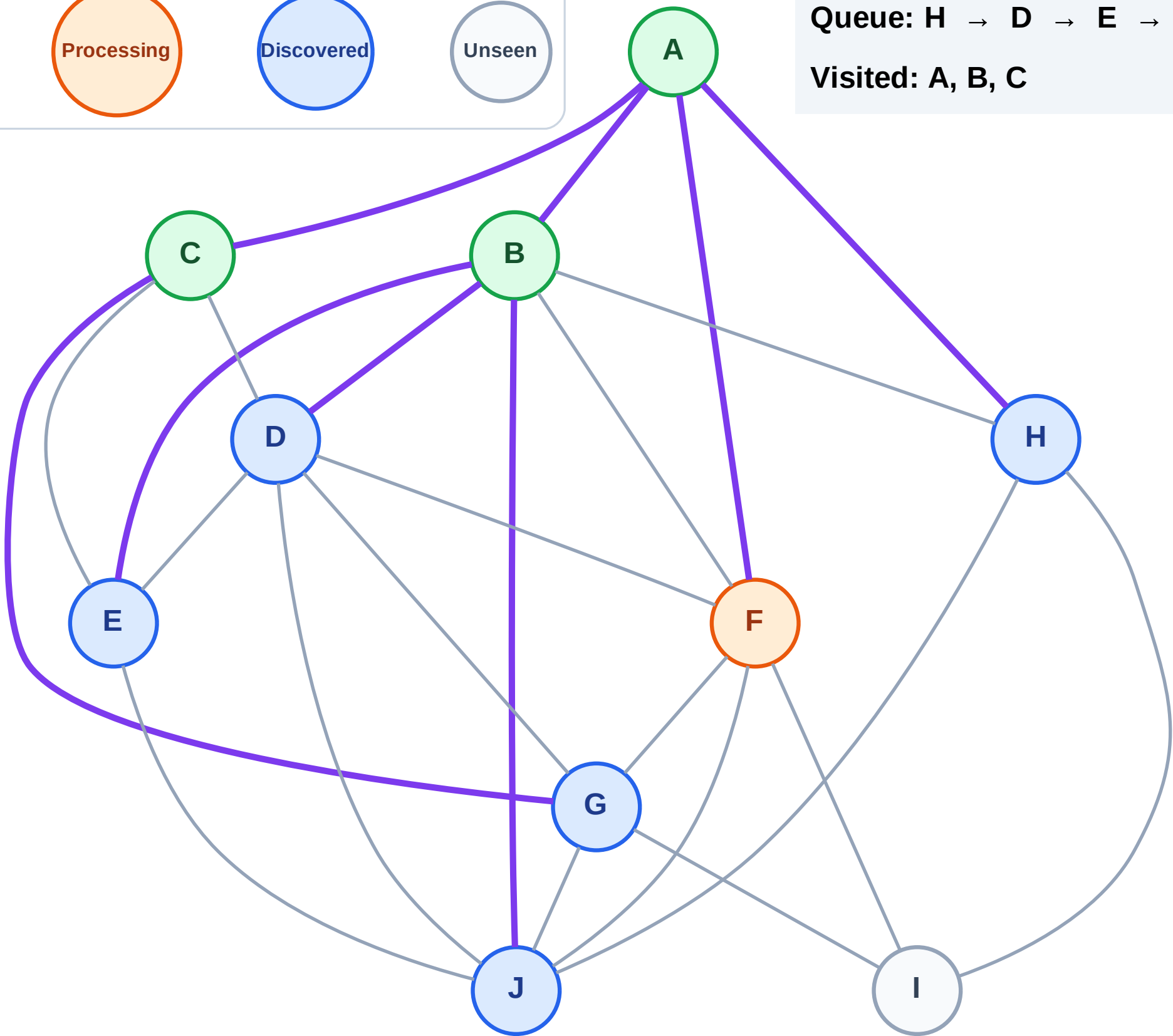
Processing

Discovered

Unseen

Queue: H → D → E → J → G

Visited: A, B, C



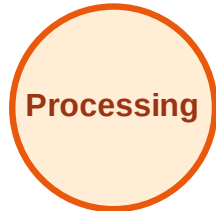
BFS Traversal

Step 9: Discover I from F

Legend



Complete



Processing



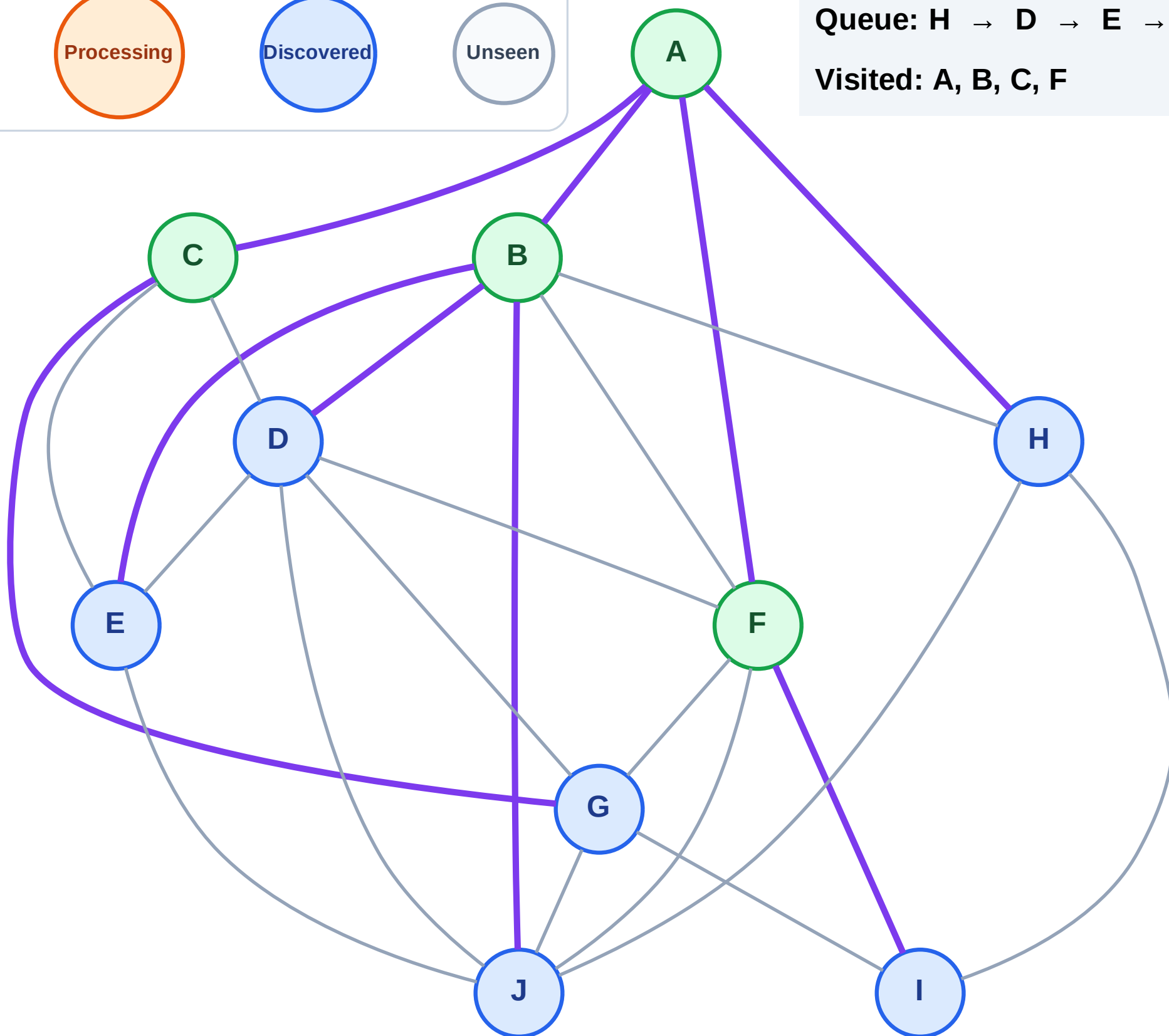
Discovered



Unseen

Queue: H → D → E → J → G → I

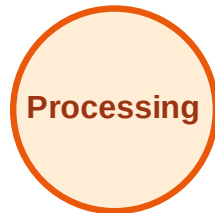
Visited: A, B, C, F



BFS Traversal

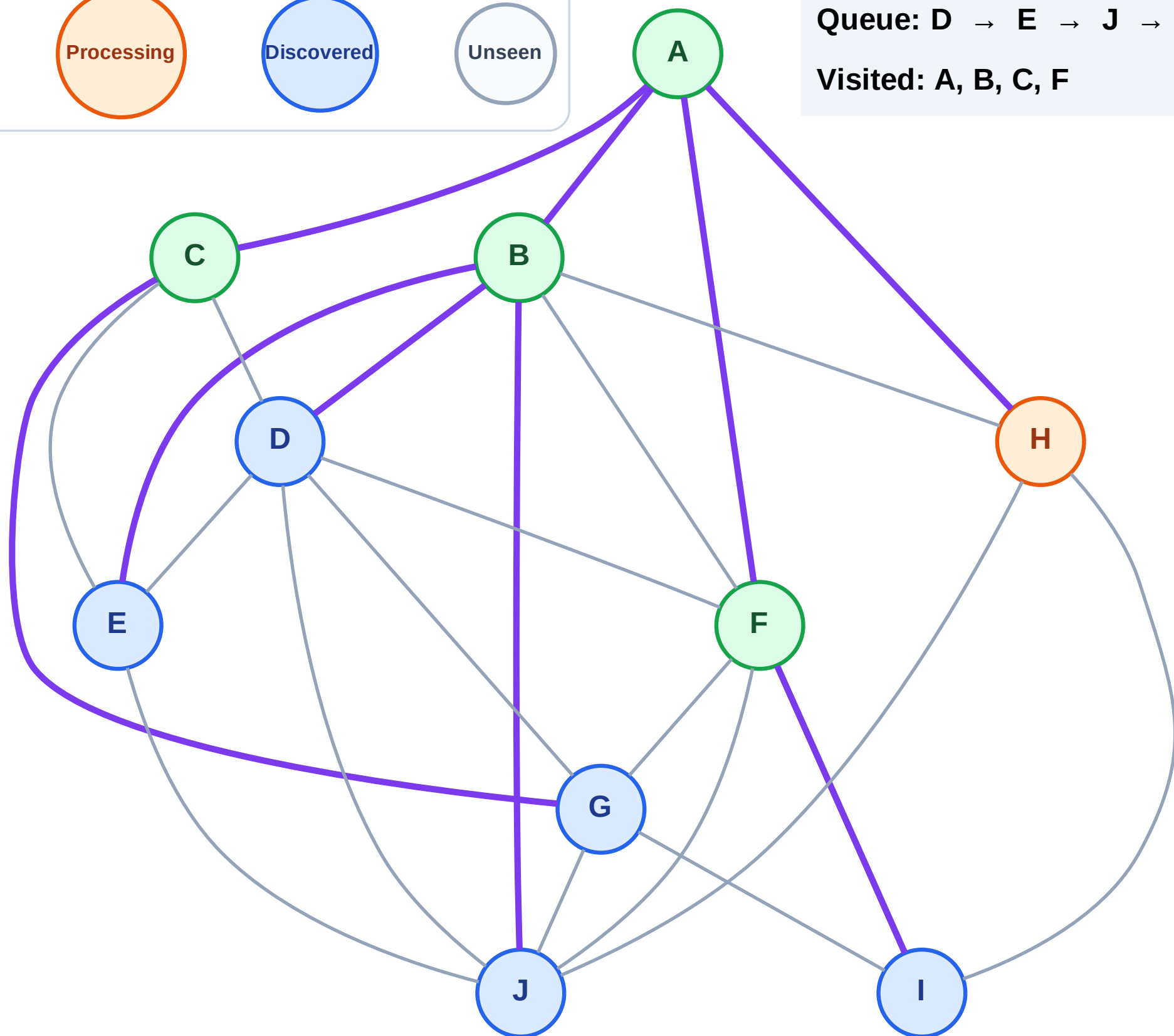
Step 10: Process node H

Legend



Queue: D → E → J → G → I

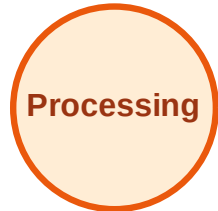
Visited: A, B, C, F



BFS Traversal

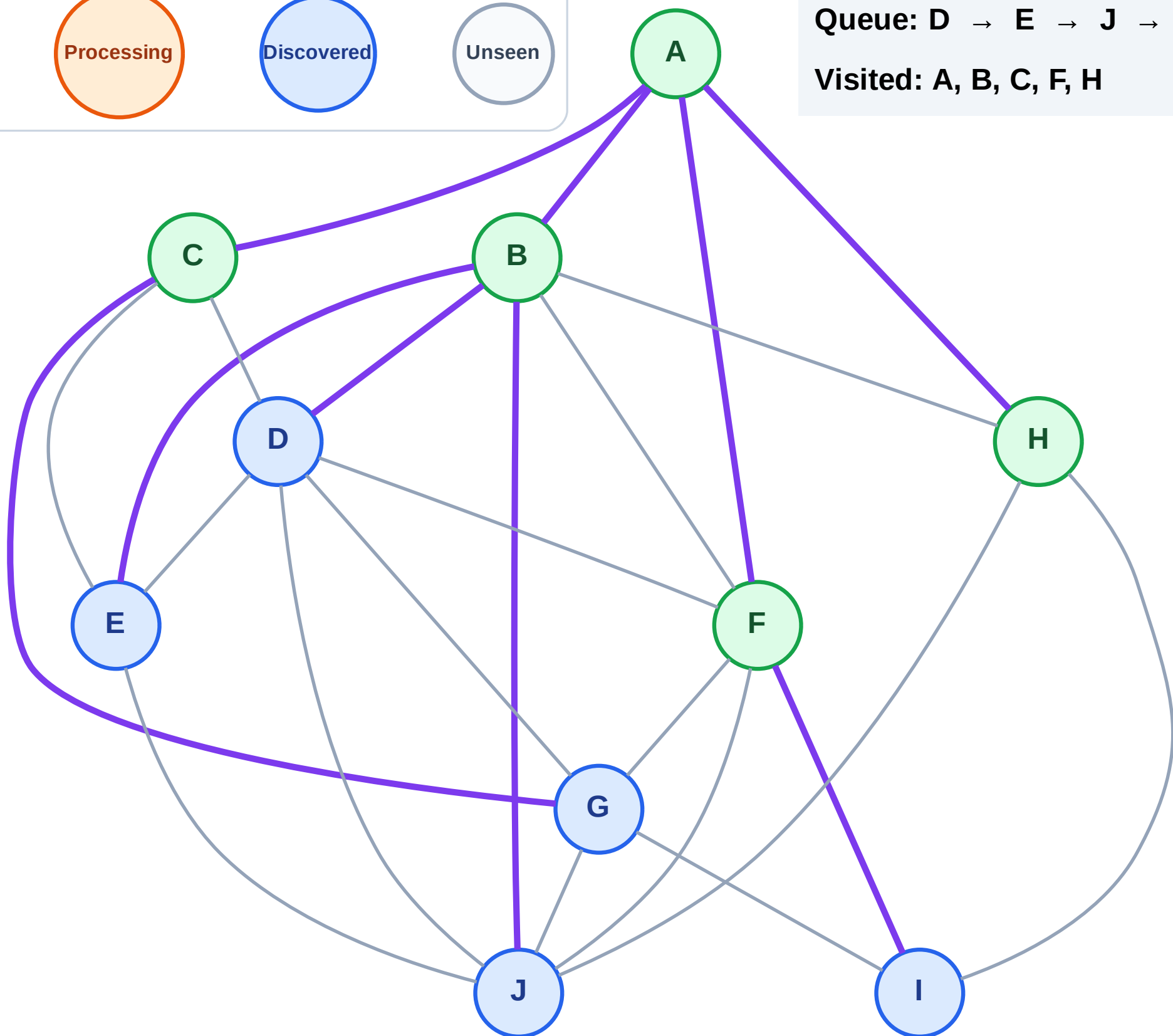
Step 11: No new neighbors from H

Legend



Queue: D → E → J → G → I

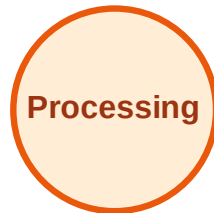
Visited: A, B, C, F, H



BFS Traversal

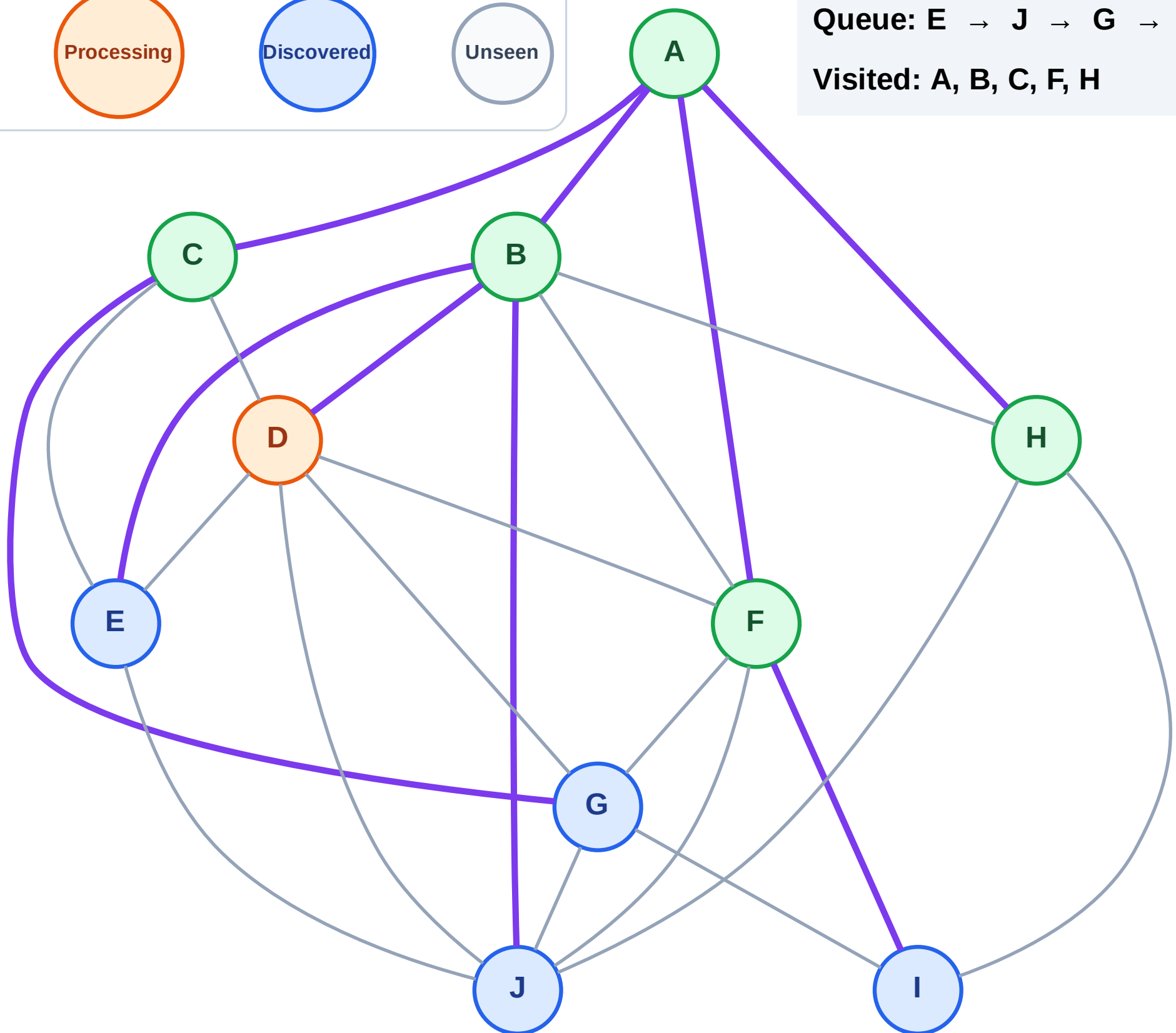
Step 12: Process node D

Legend



Queue: E → J → G → I

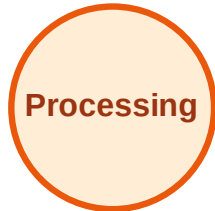
Visited: A, B, C, F, H



BFS Traversal

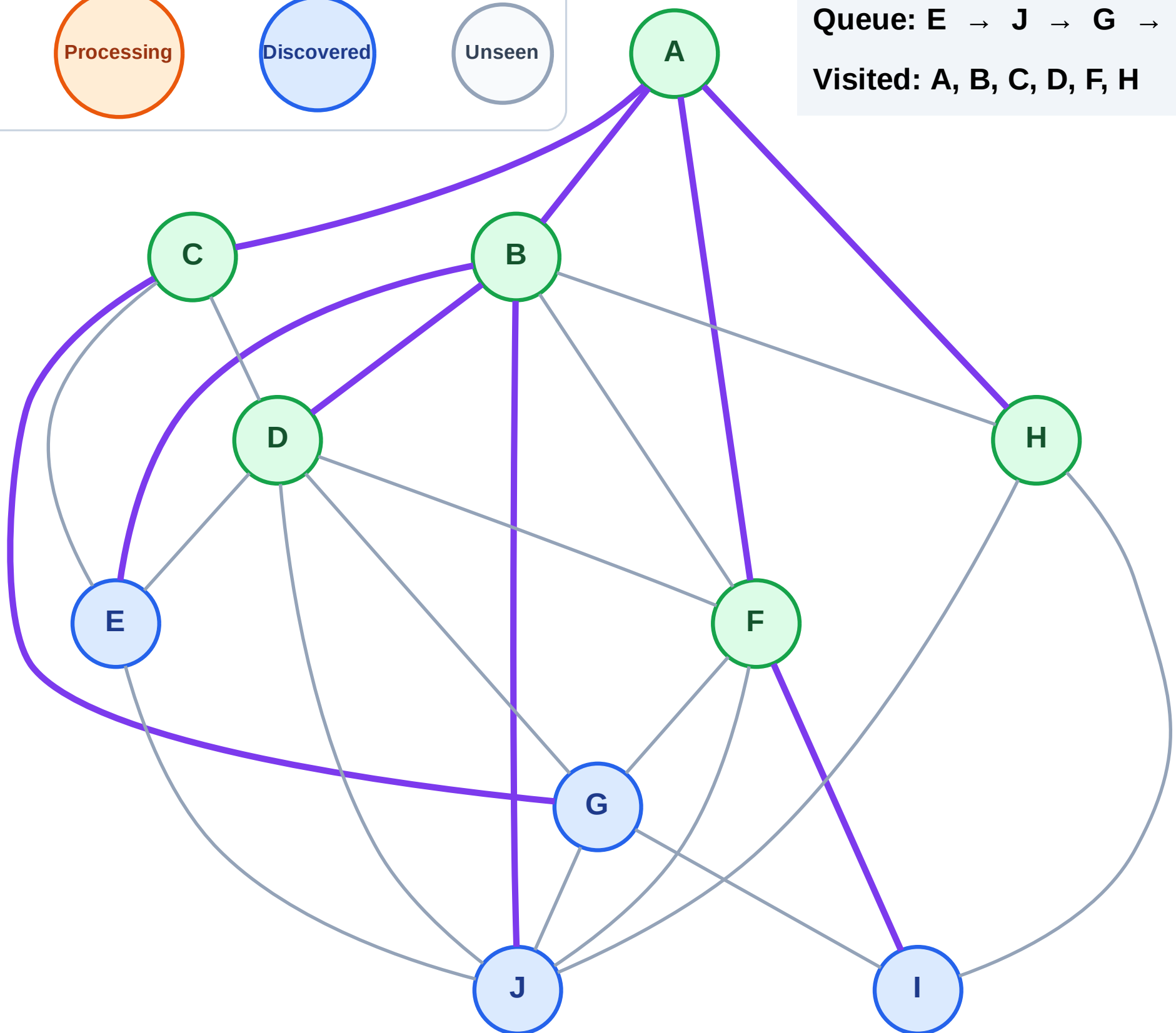
Step 13: No new neighbors from D

Legend



Queue: E → J → G → I

Visited: A, B, C, D, F, H



BFS Traversal

Step 14: Process node E

Legend

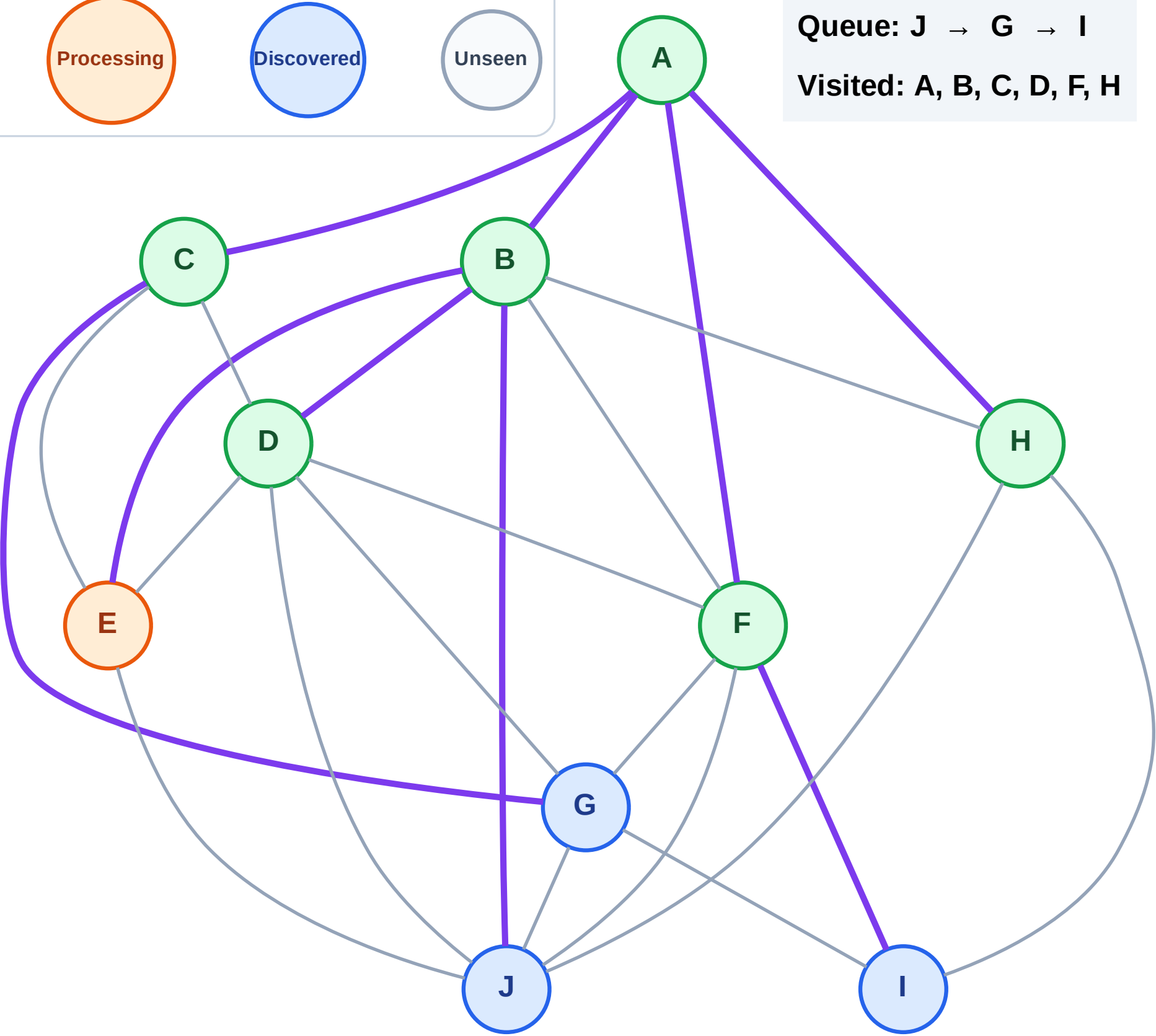
Complete

Processing

Discovered

Unseen

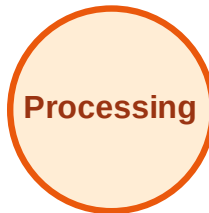
Queue: J → G → I
Visited: A, B, C, D, F, H



BFS Traversal

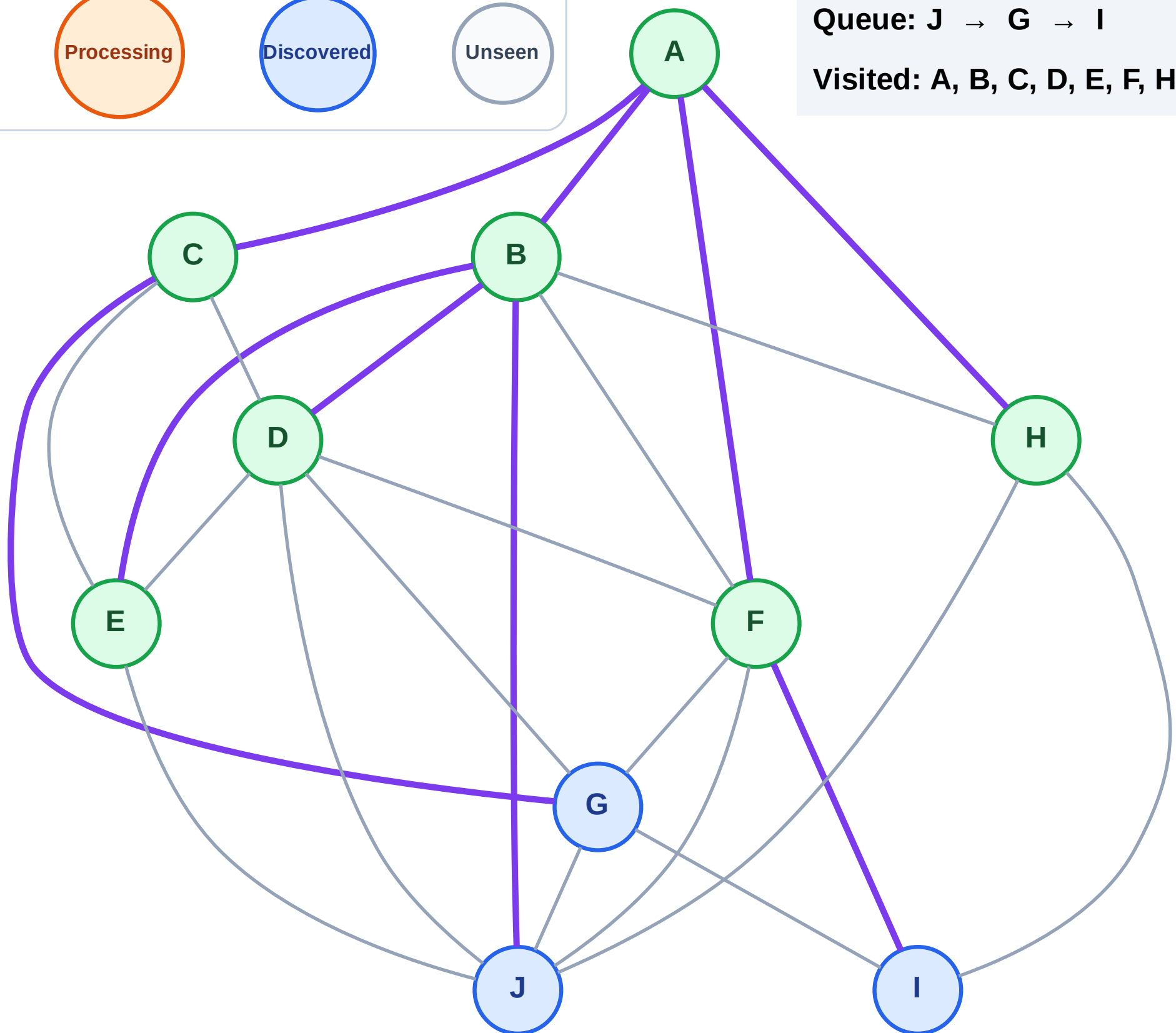
Step 15: No new neighbors from E

Legend



Queue: J → G → I

Visited: A, B, C, D, E, F, H



BFS Traversal

Step 16: Process node J

Legend

Complete

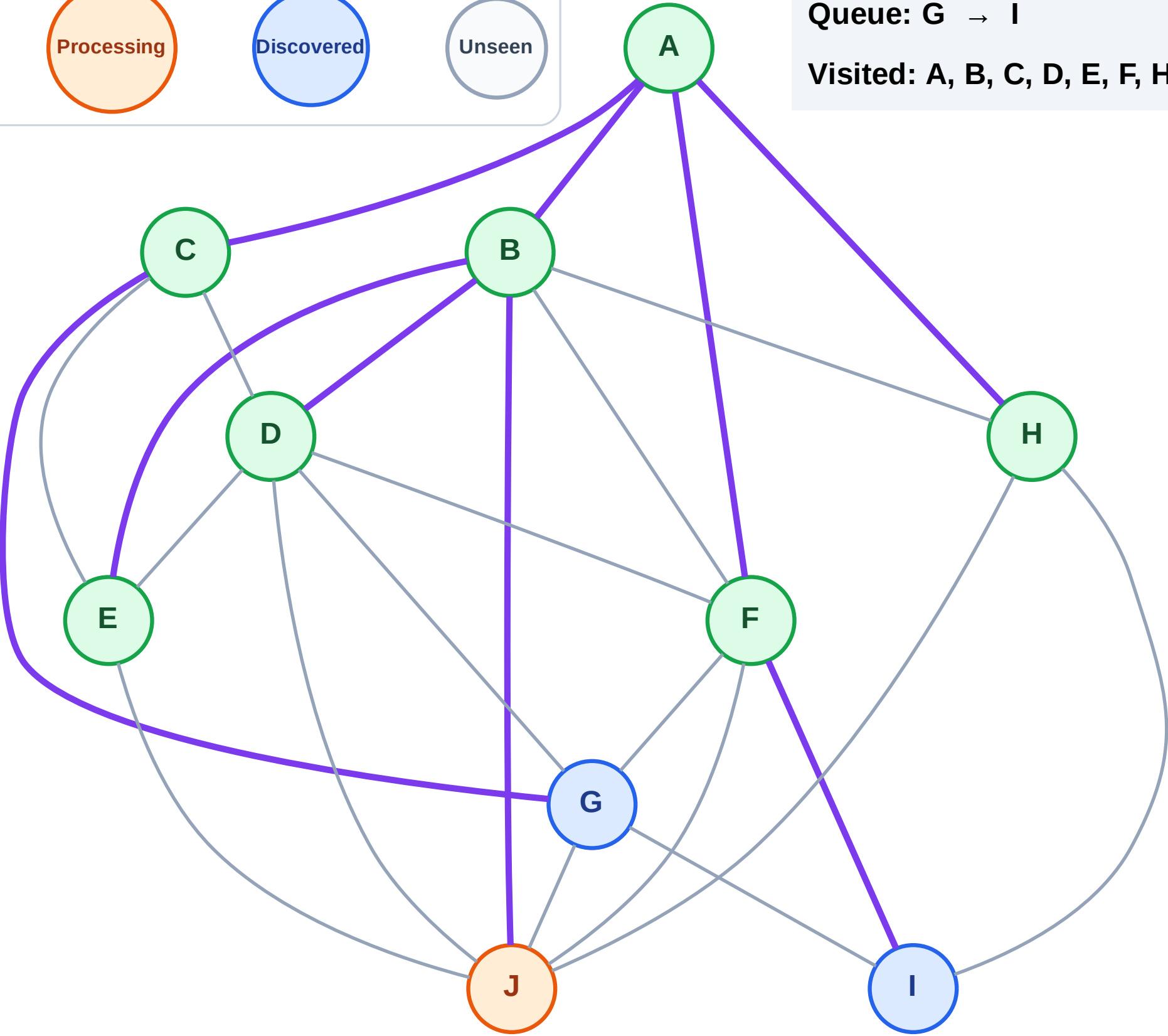
Processing

Discovered

Unseen

Queue: G → I

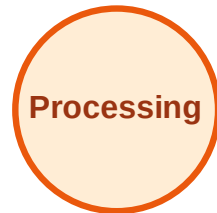
Visited: A, B, C, D, E, F, H



BFS Traversal

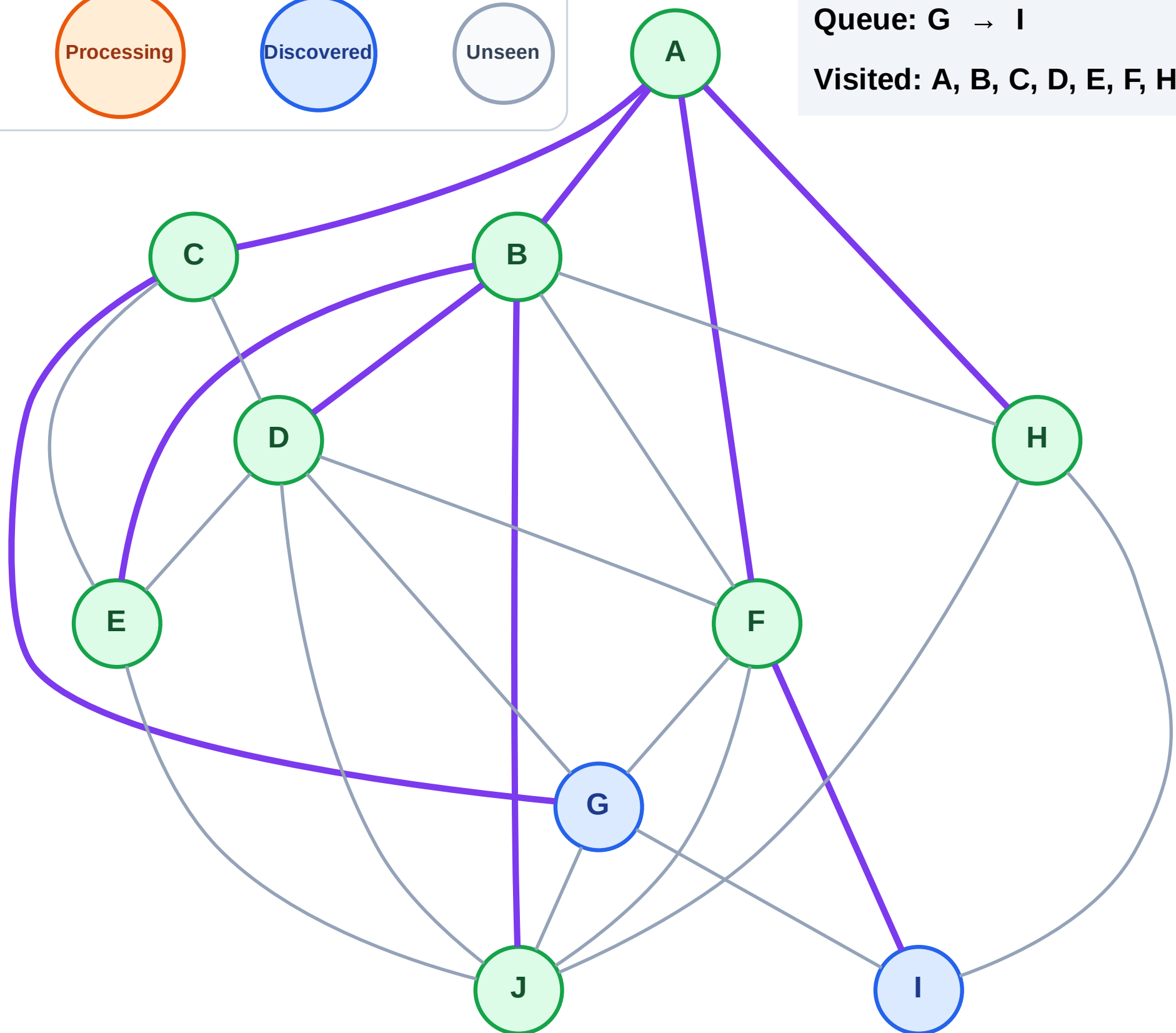
Step 17: No new neighbors from J

Legend



Queue: G → I

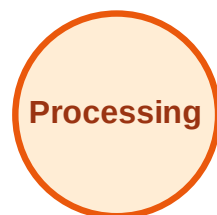
Visited: A, B, C, D, E, F, H, J



BFS Traversal

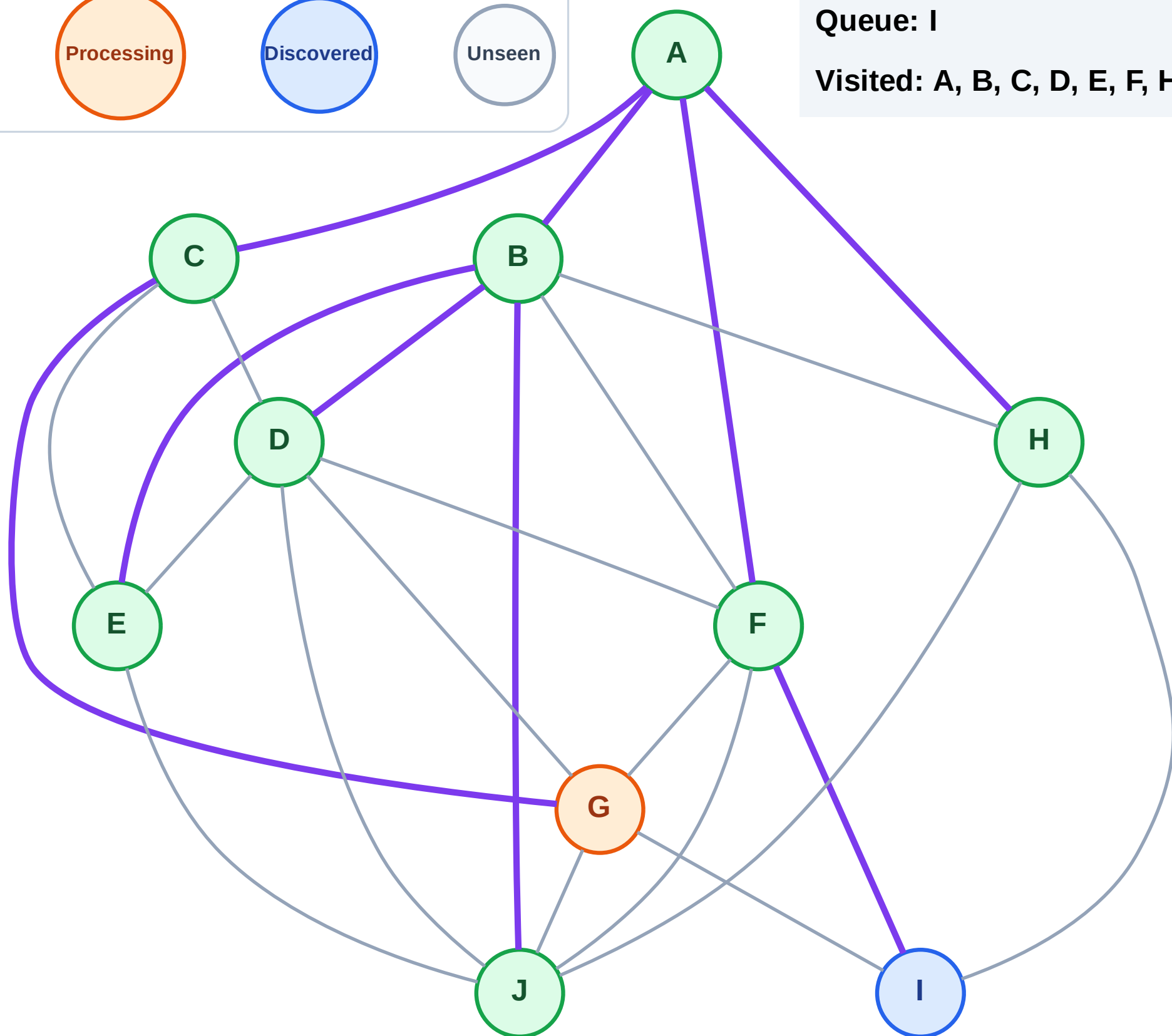
Step 18: Process node G

Legend



Queue: I

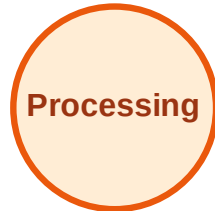
Visited: A, B, C, D, E, F, H, J



BFS Traversal

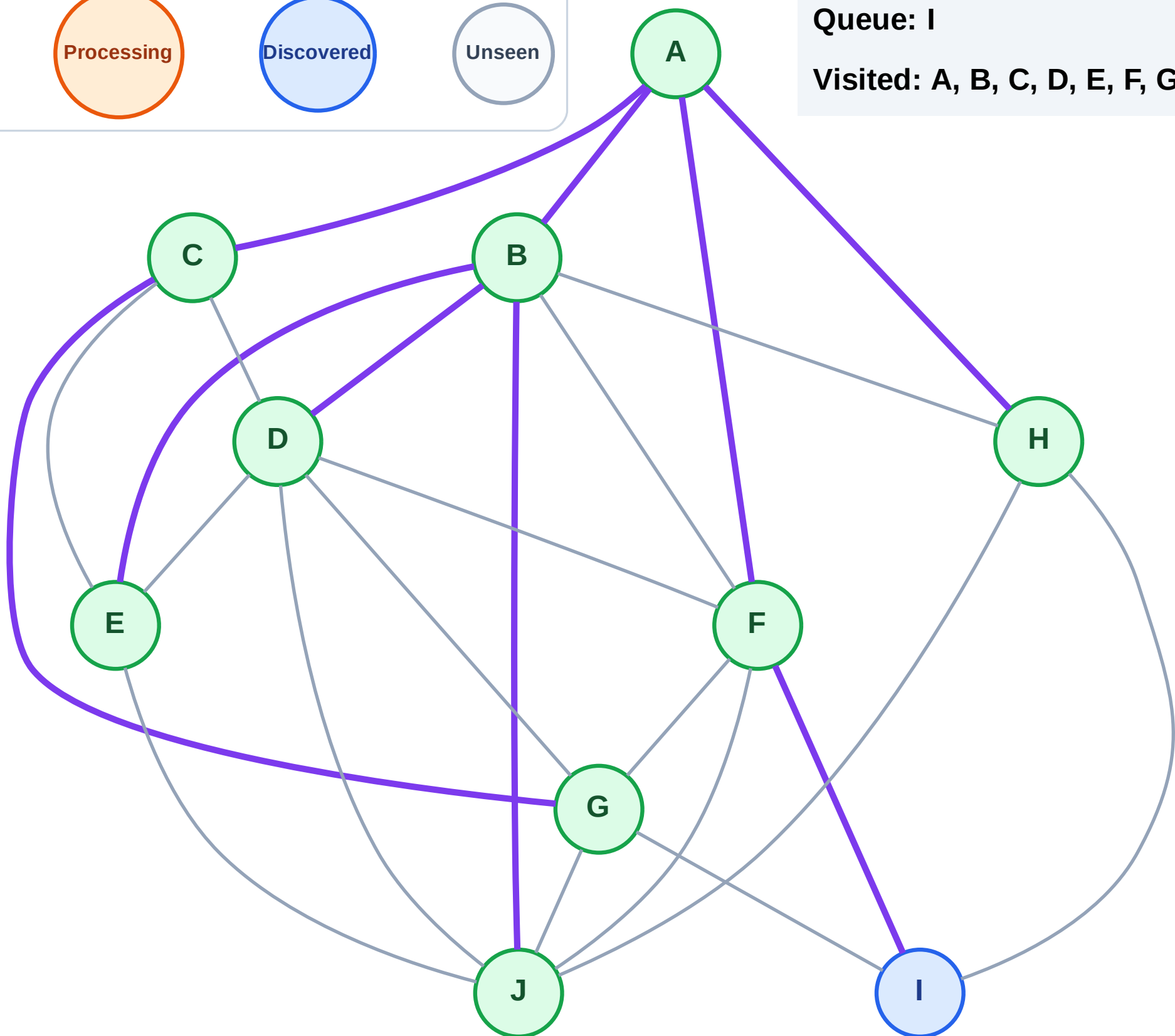
Step 19: No new neighbors from G

Legend



Queue: I

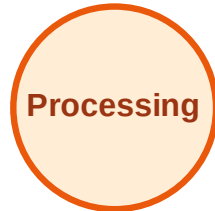
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

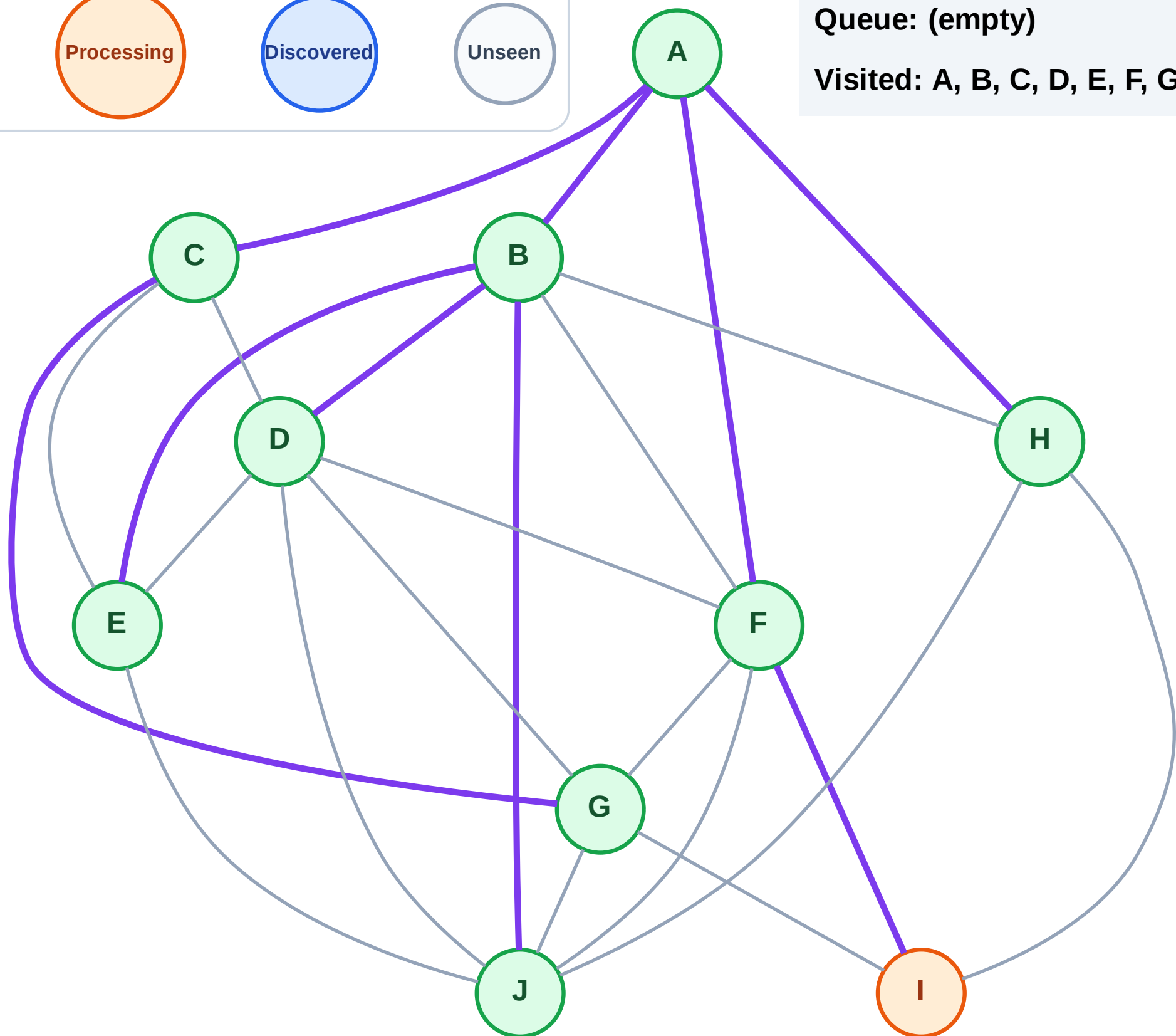
Step 20: Process node I

Legend



Queue: (empty)

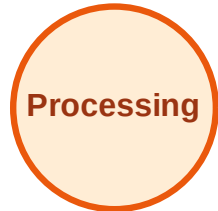
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

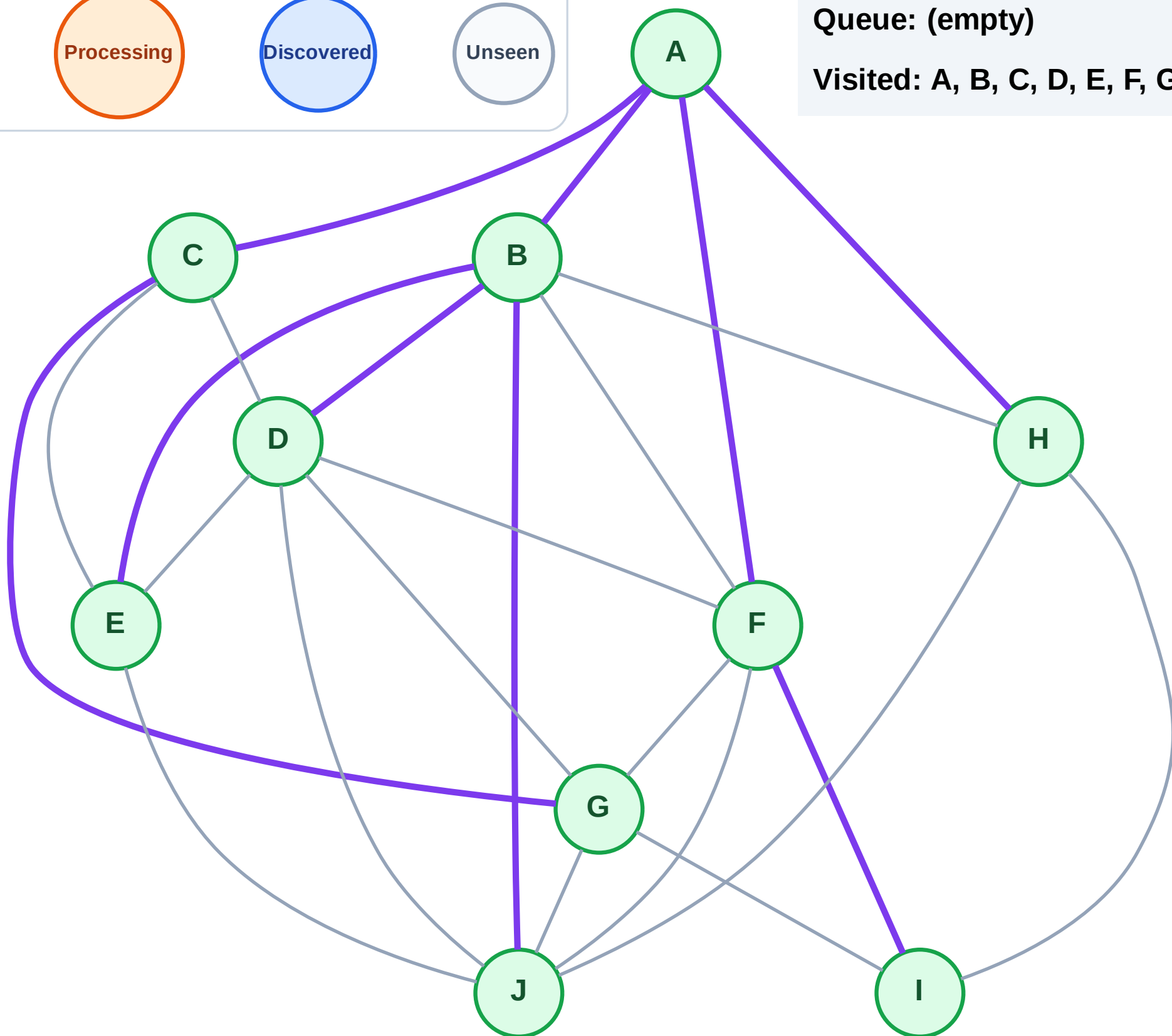
Step 21: No new neighbors from I

Legend



Queue: (empty)

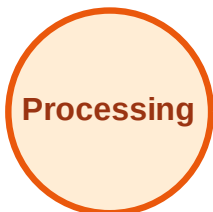
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

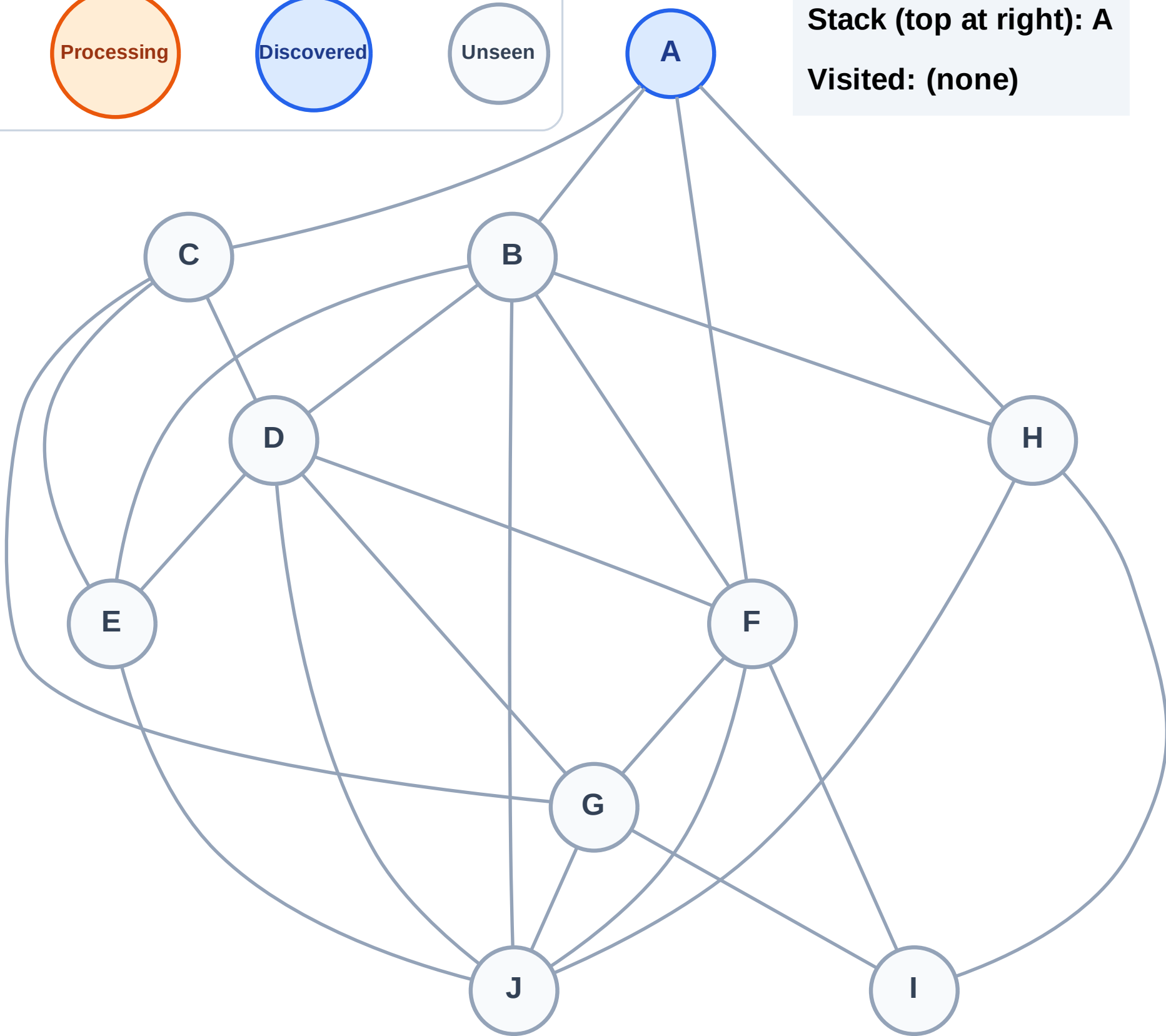
Step 1: Start at A

Legend



Stack (top at right): A

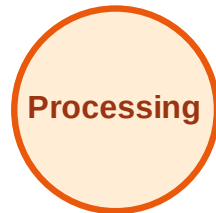
Visited: (none)



DFS Traversal

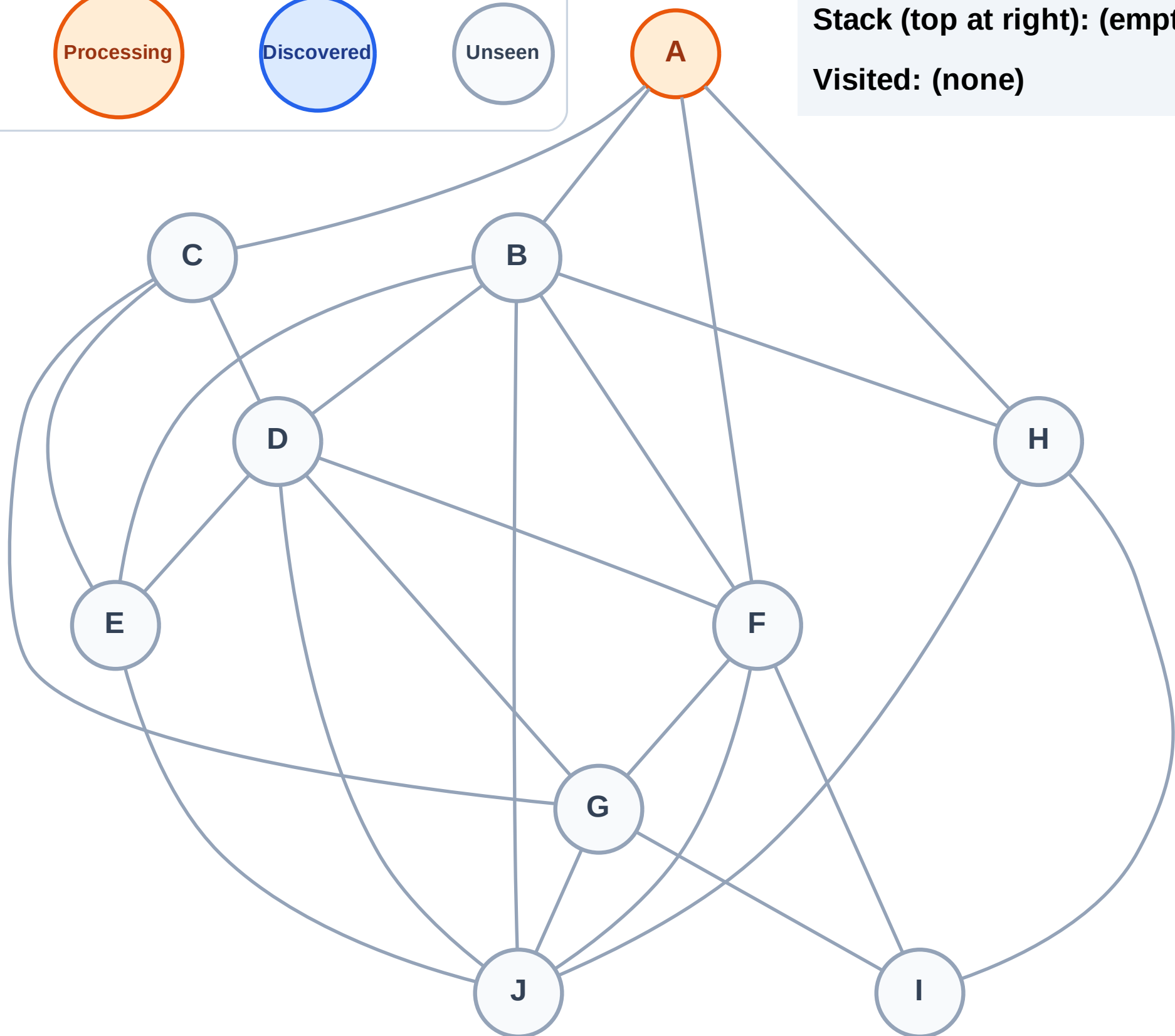
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover H, F, C, B from A

Legend

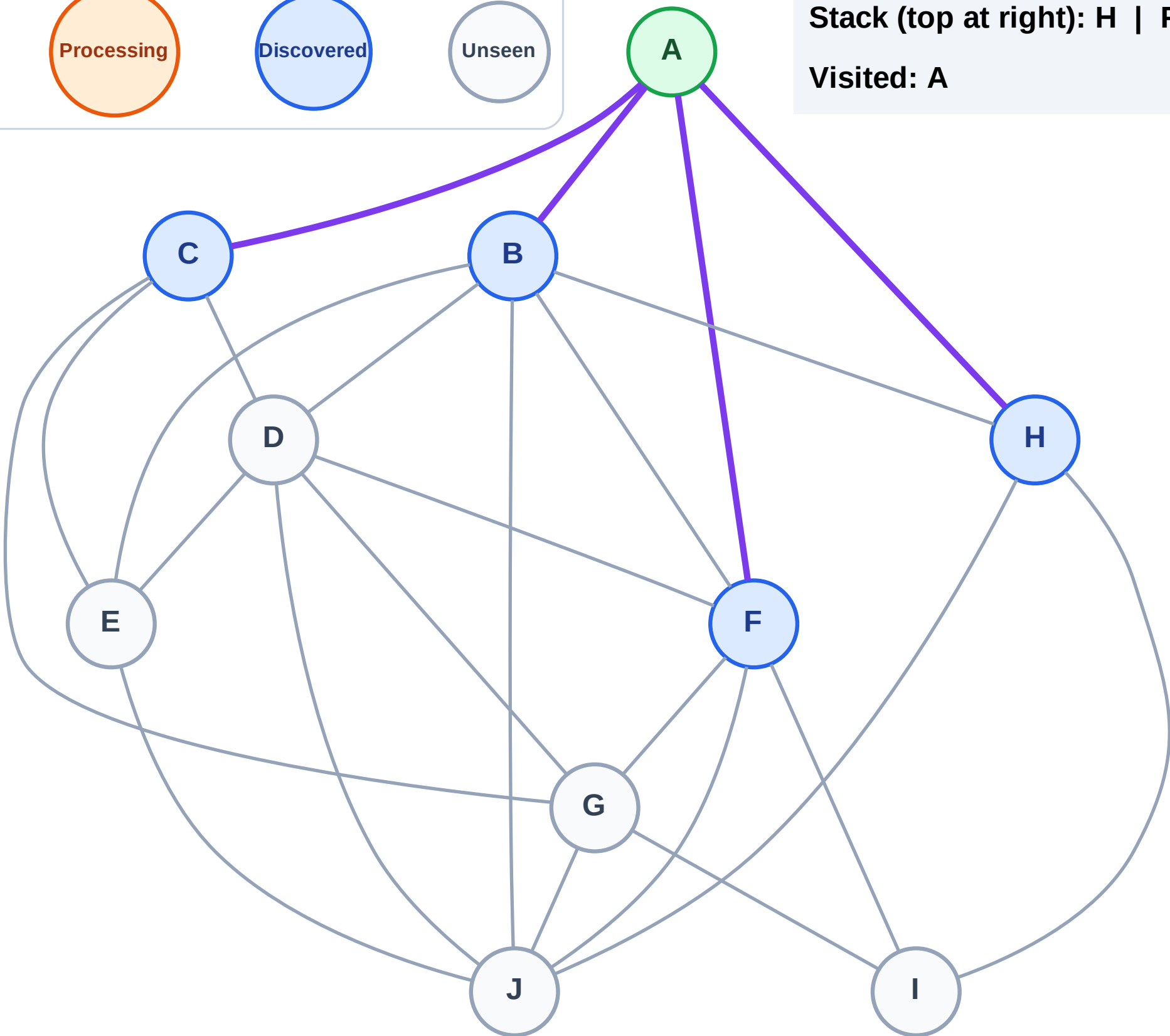
Complete

Processing

Discovered

Unseen

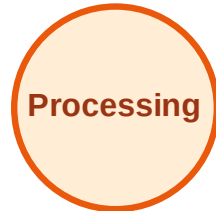
Stack (top at right): H | F | C | B
Visited: A



DFS Traversal

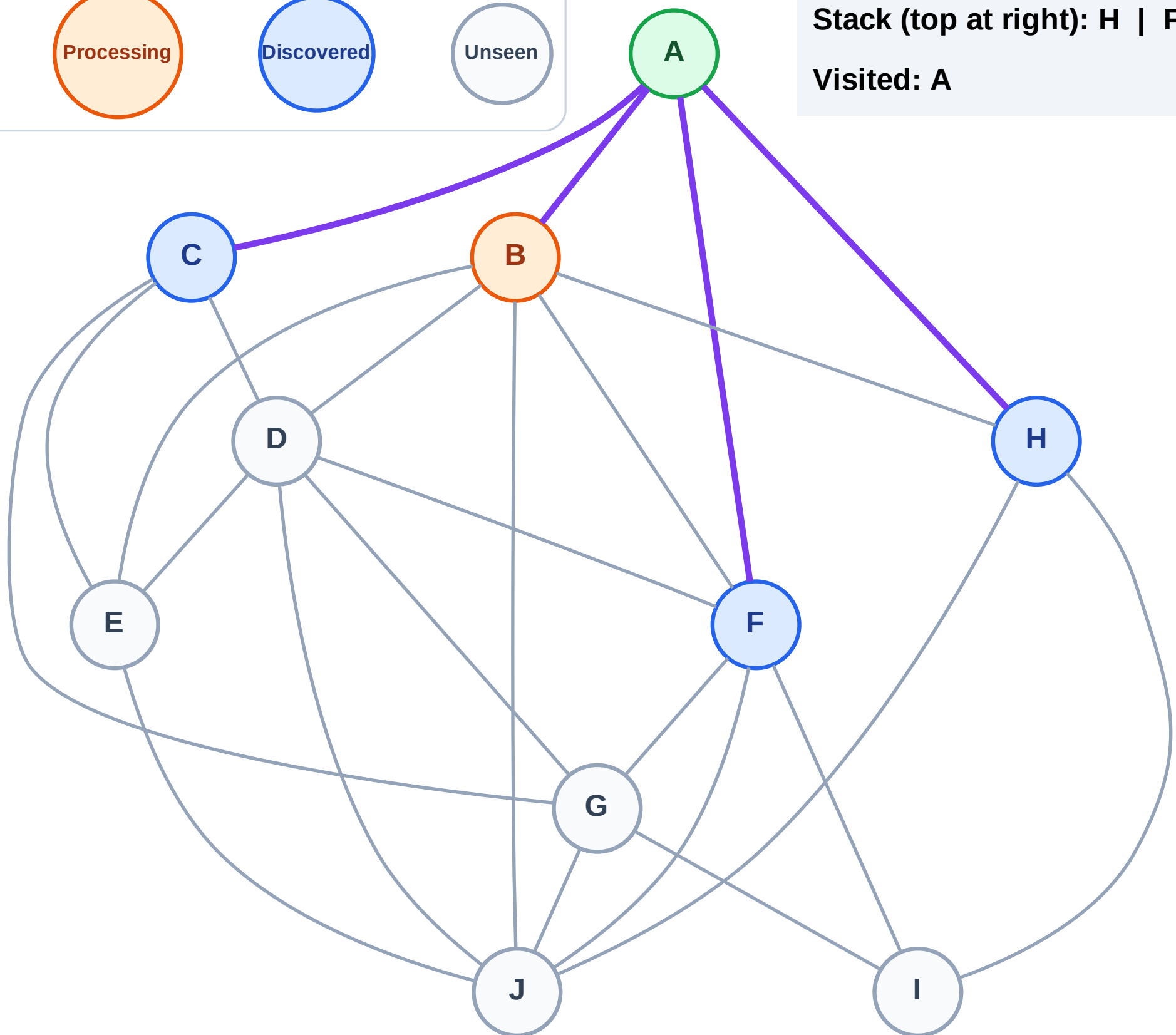
Step 4: Process node B

Legend



Stack (top at right): H | F | C

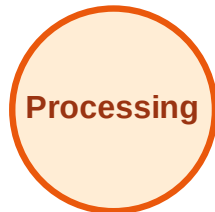
Visited: A



DFS Traversal

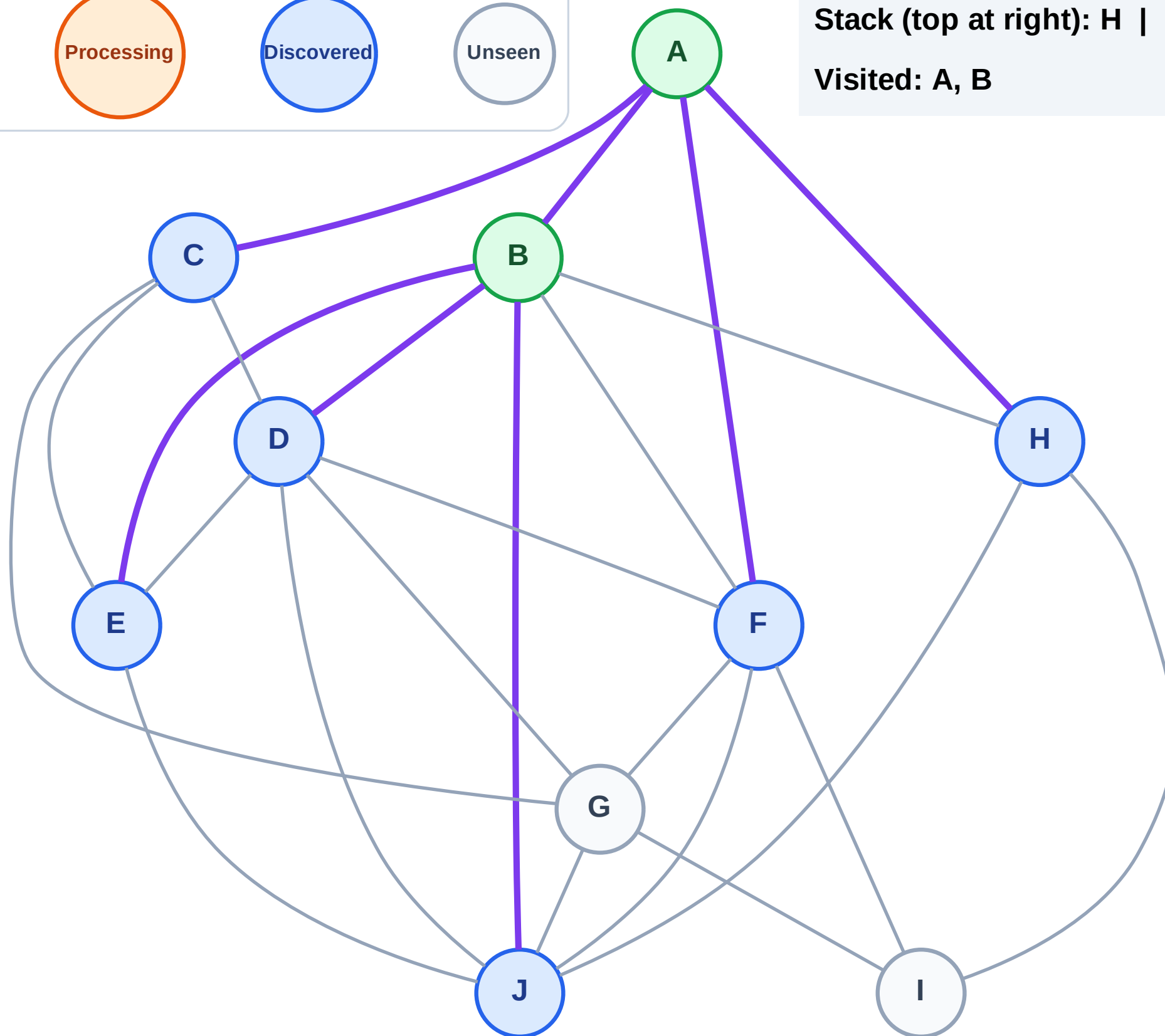
Step 5: Discover J, E, D from B

Legend



Stack (top at right): H | F | C | J | E | D

Visited: A, B



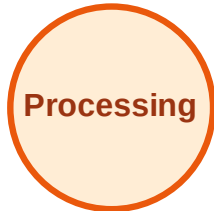
DFS Traversal

Step 6: Process node D

Legend



Complete



Processing



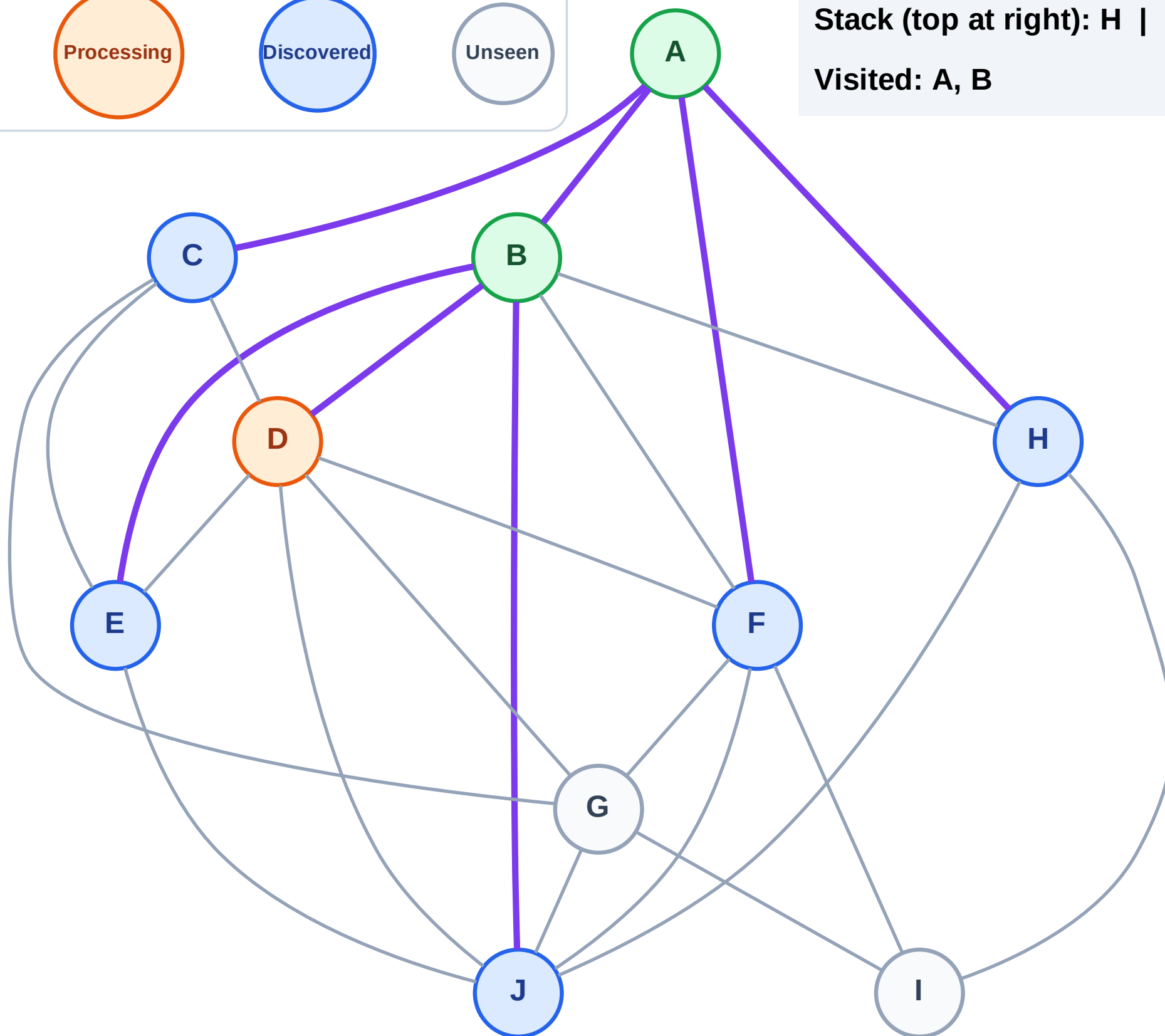
Discovered



Unseen

Stack (top at right): H | F | C | J | E

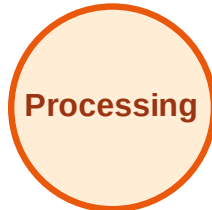
Visited: A, B



DFS Traversal

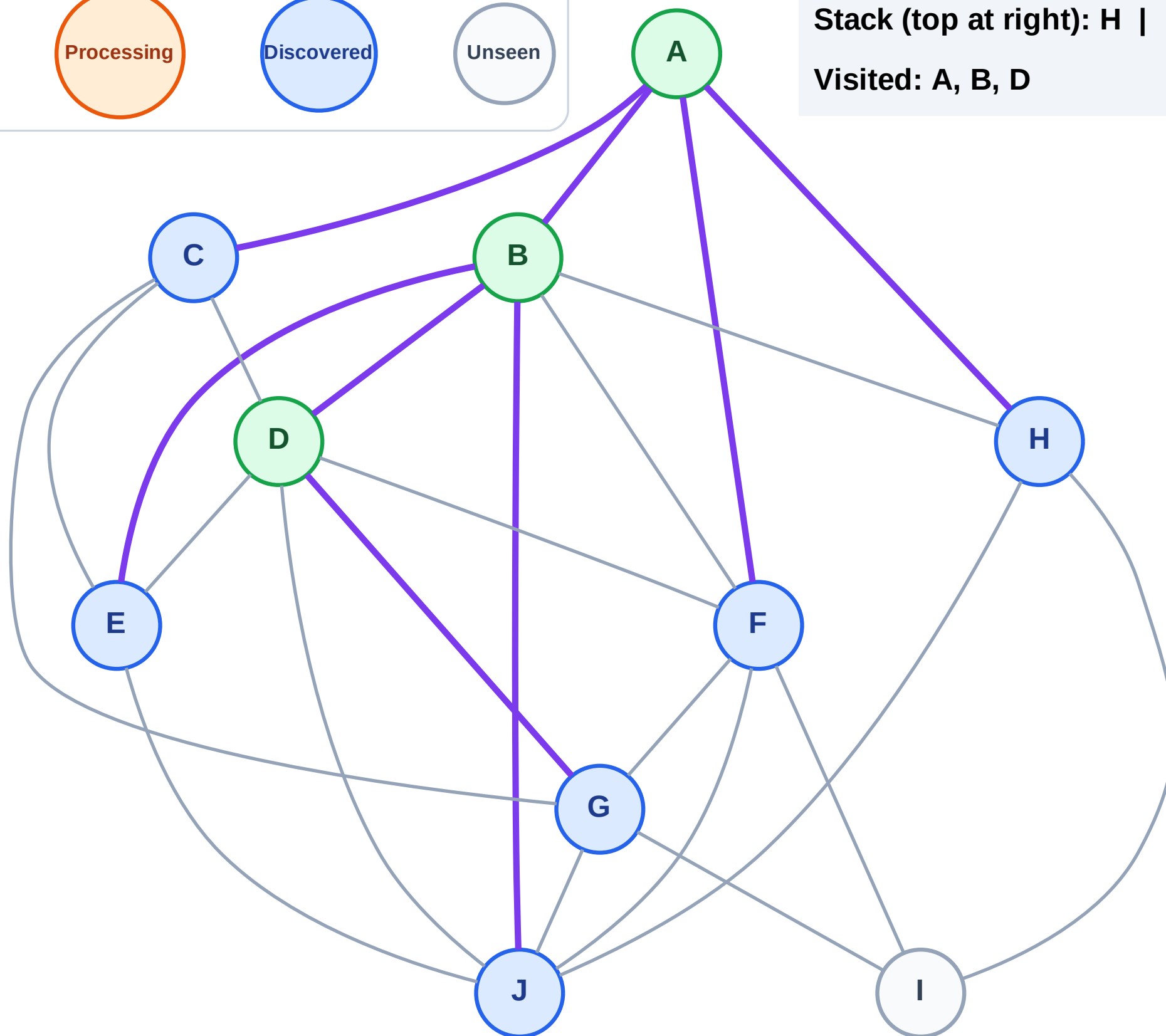
Step 7: Discover G from D

Legend



Stack (top at right): H | F | C | J | E | G

Visited: A, B, D



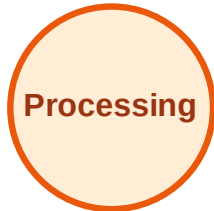
DFS Traversal

Step 8: Process node G

Legend



Complete



Processing



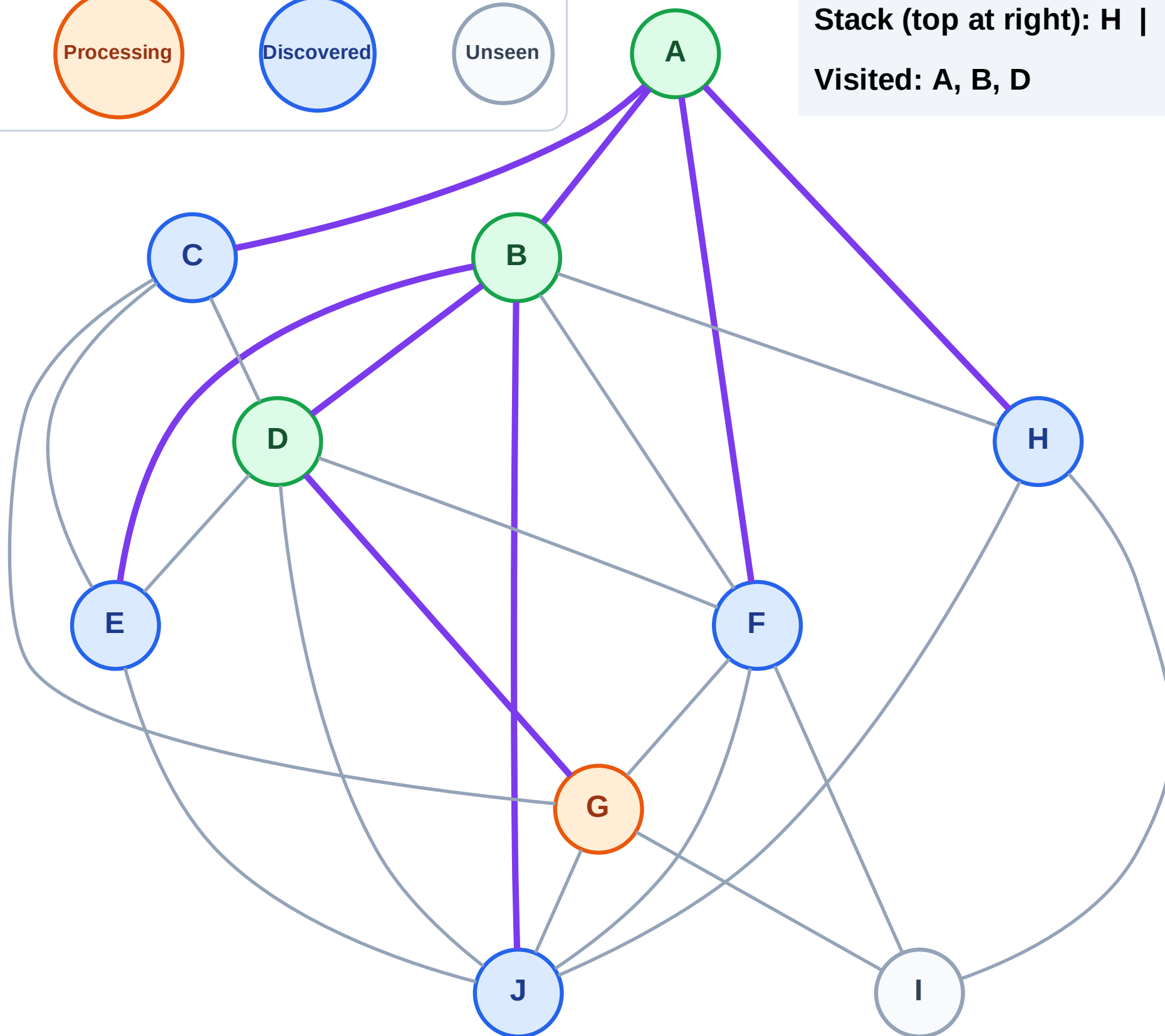
Discovered



Unseen

Stack (top at right): H | F | C | J | E

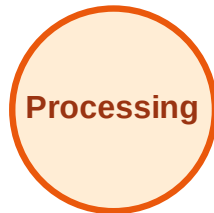
Visited: A, B, D



DFS Traversal

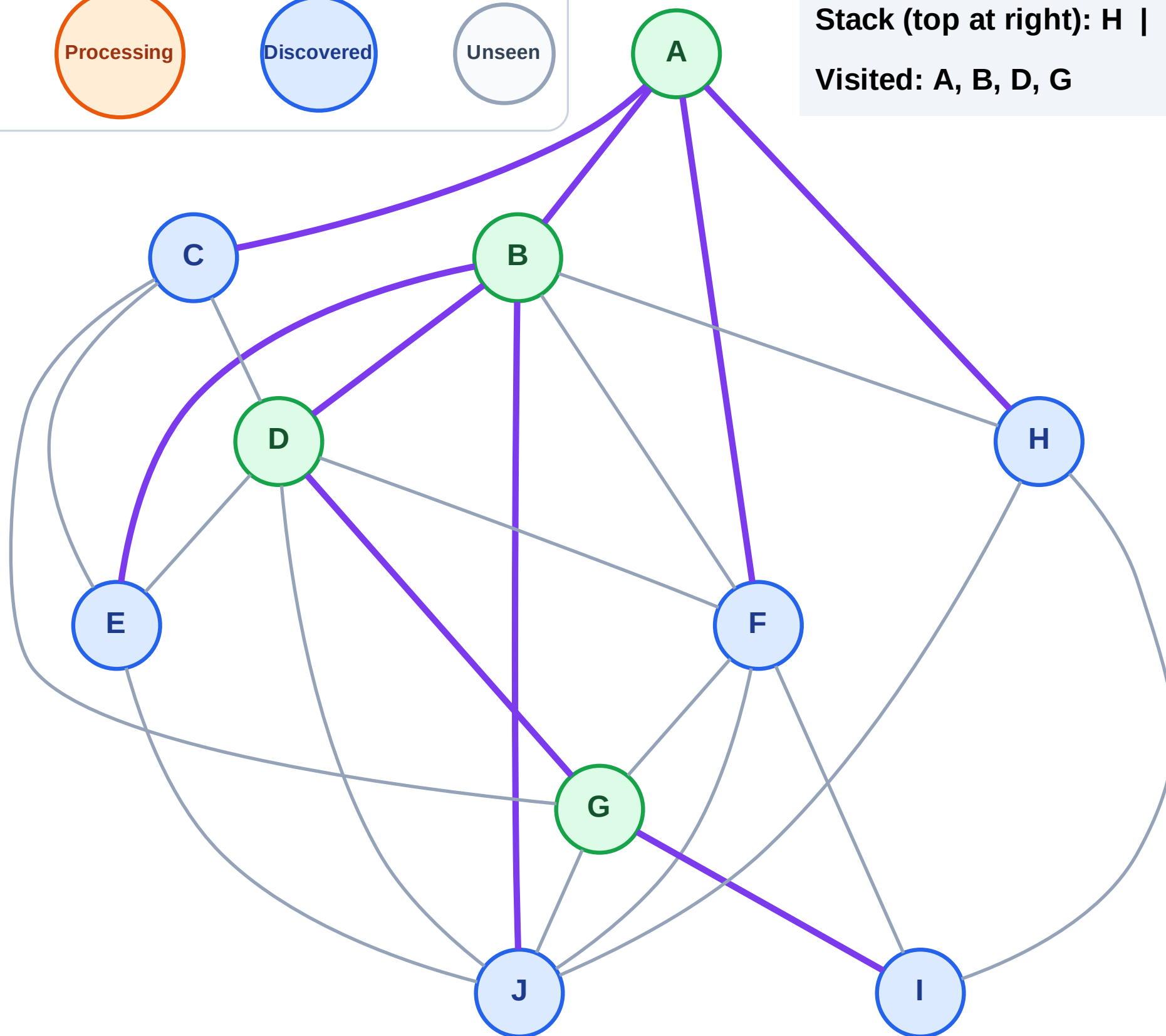
Step 9: Discover I from G

Legend



Stack (top at right): H | F | C | J | E | I

Visited: A, B, D, G



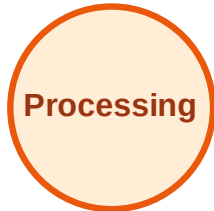
DFS Traversal

Step 10: Process node I

Legend



Complete



Processing



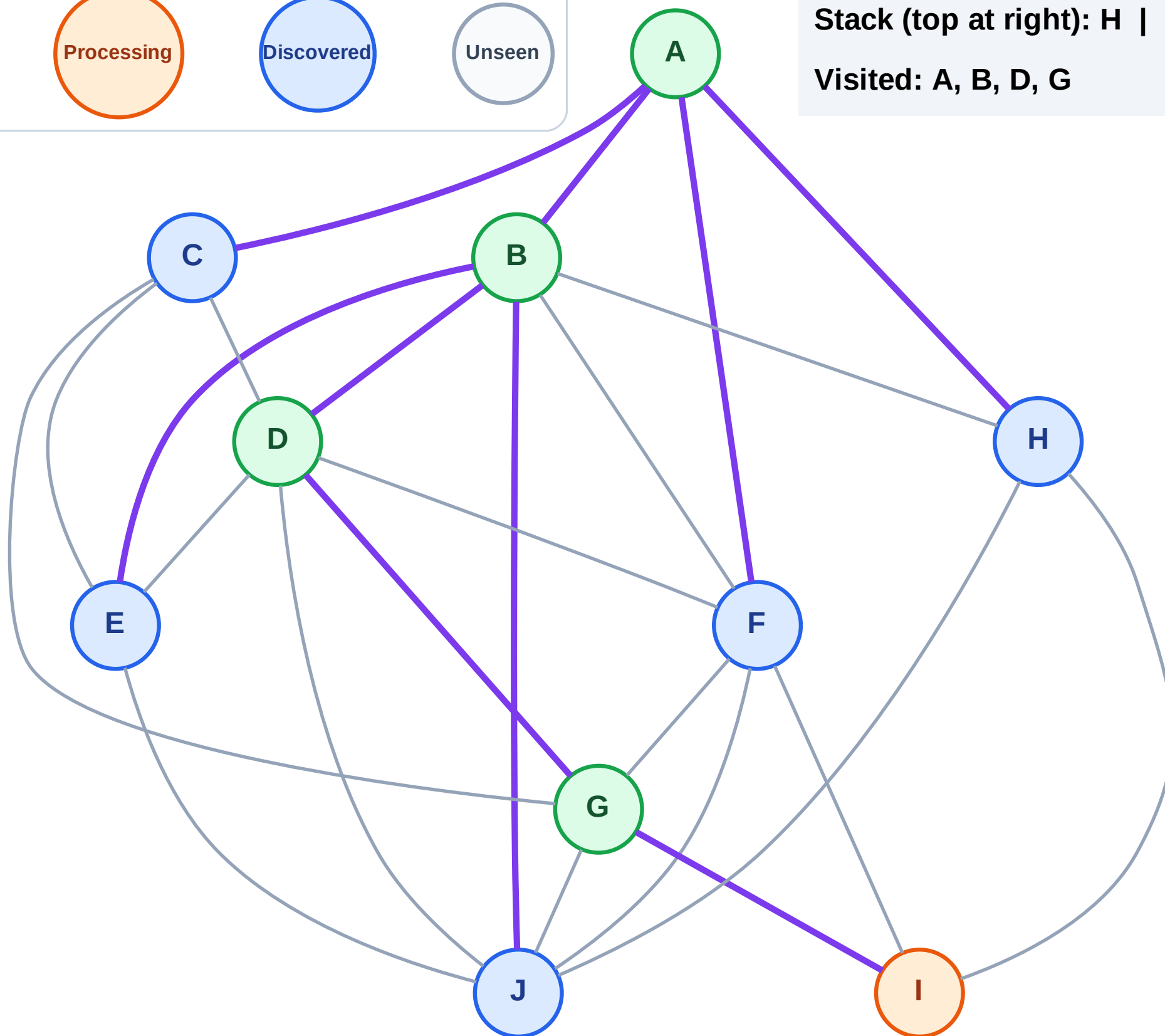
Discovered



Unseen

Stack (top at right): H | F | C | J | E

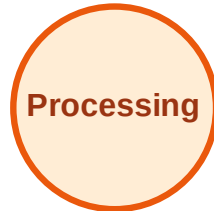
Visited: A, B, D, G



DFS Traversal

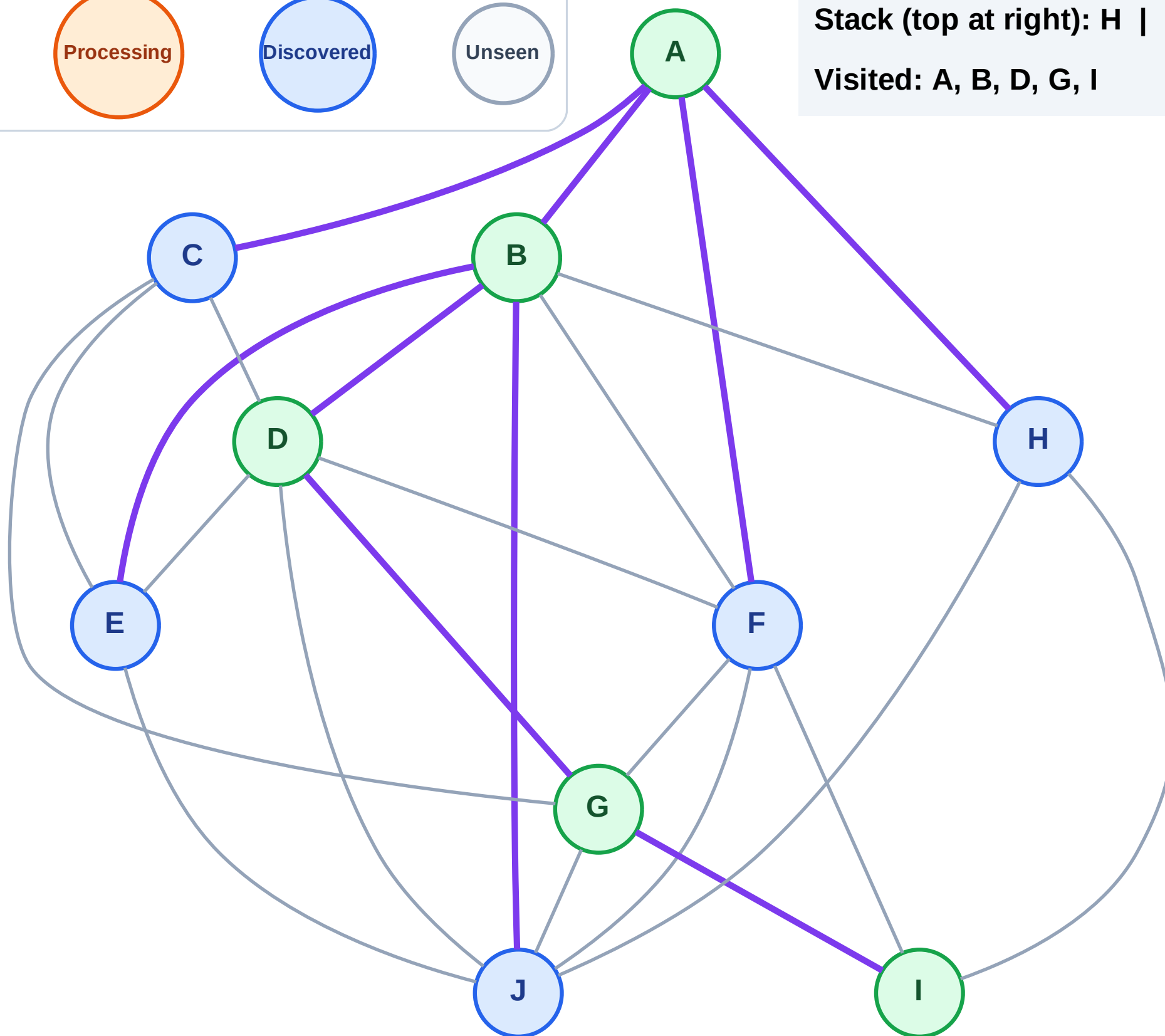
Step 11: No new neighbors from I

Legend



Stack (top at right): H | F | C | J | E

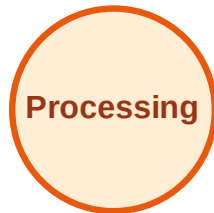
Visited: A, B, D, G, I



DFS Traversal

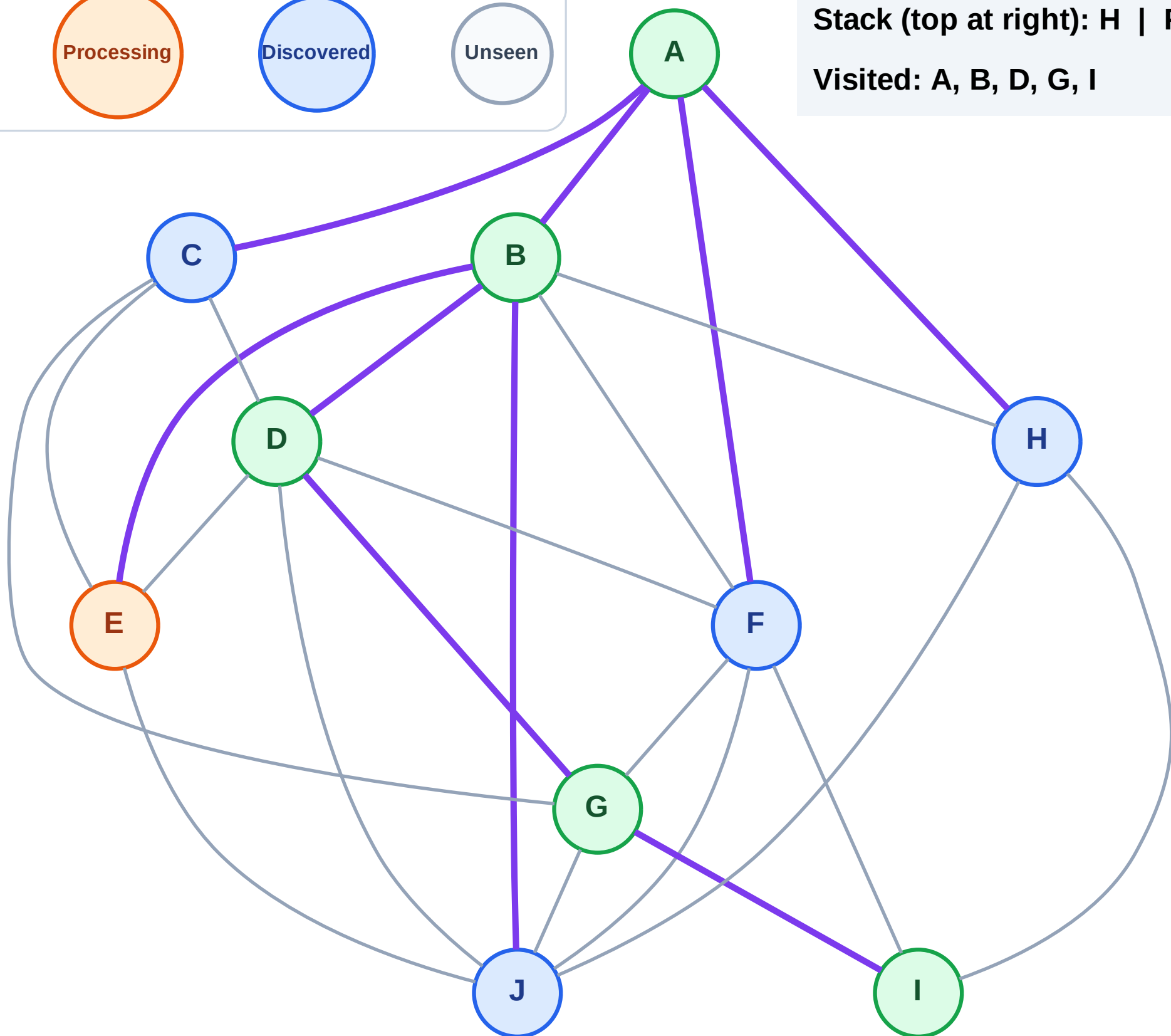
Step 12: Process node E

Legend



Stack (top at right): H | F | C | J

Visited: A, B, D, G, I



DFS Traversal

Step 13: No new neighbors from E

Legend

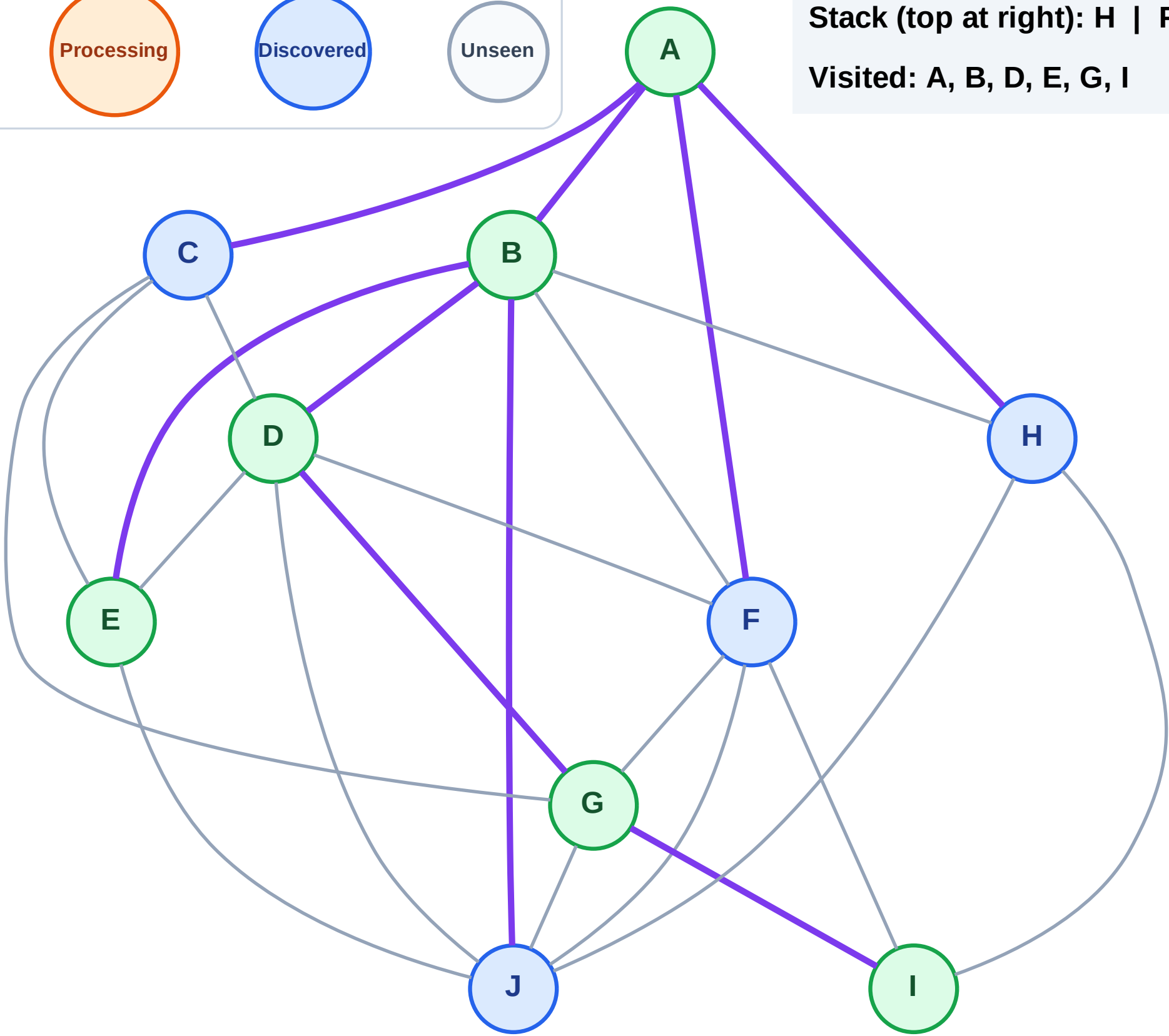
Complete

Processing

Discovered

Unseen

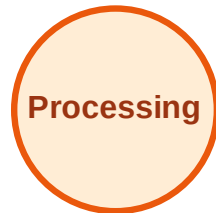
Stack (top at right): H | F | C | J
Visited: A, B, D, E, G, I



DFS Traversal

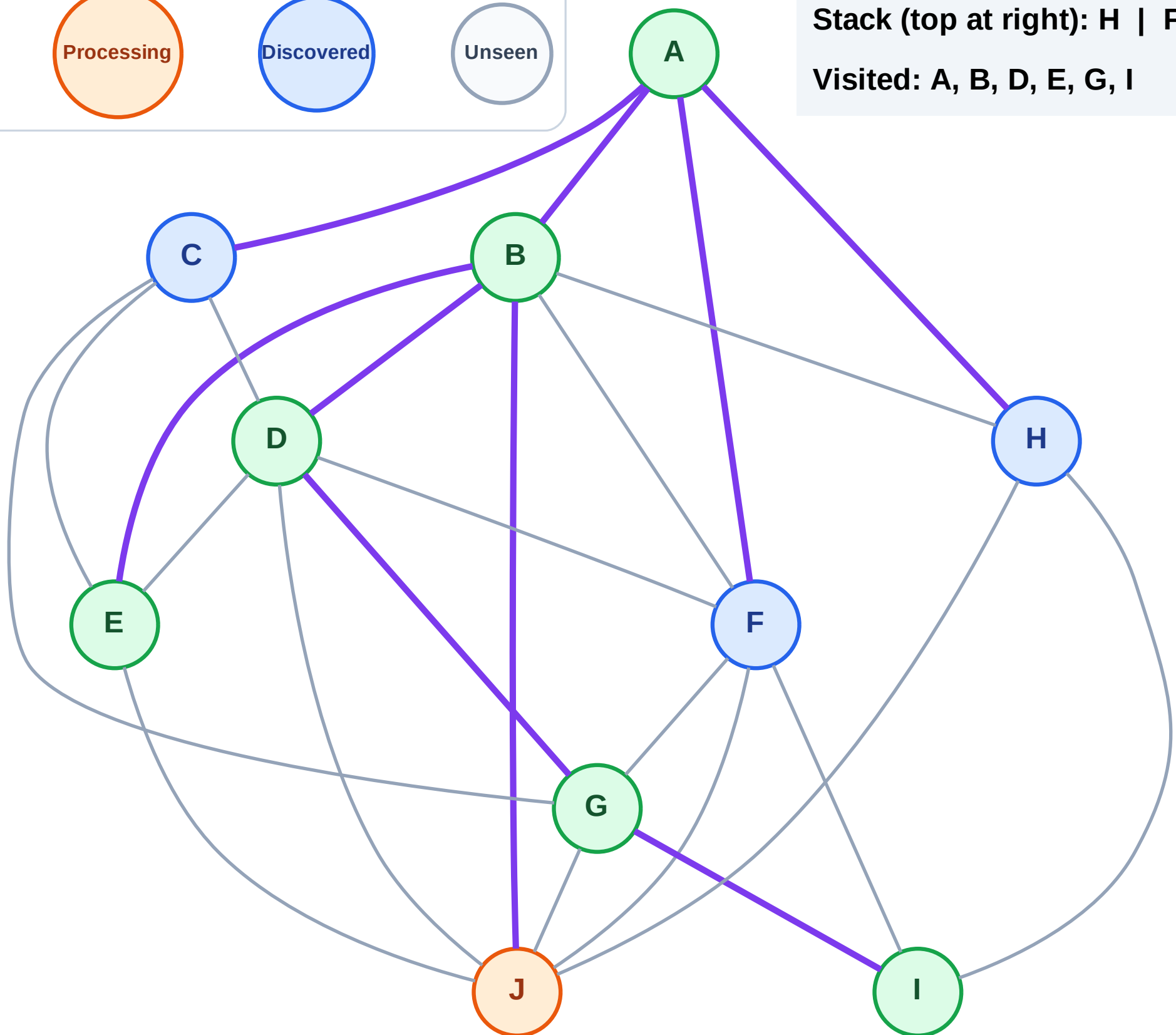
Step 14: Process node J

Legend



Stack (top at right): H | F | C

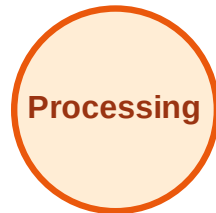
Visited: A, B, D, E, G, I



DFS Traversal

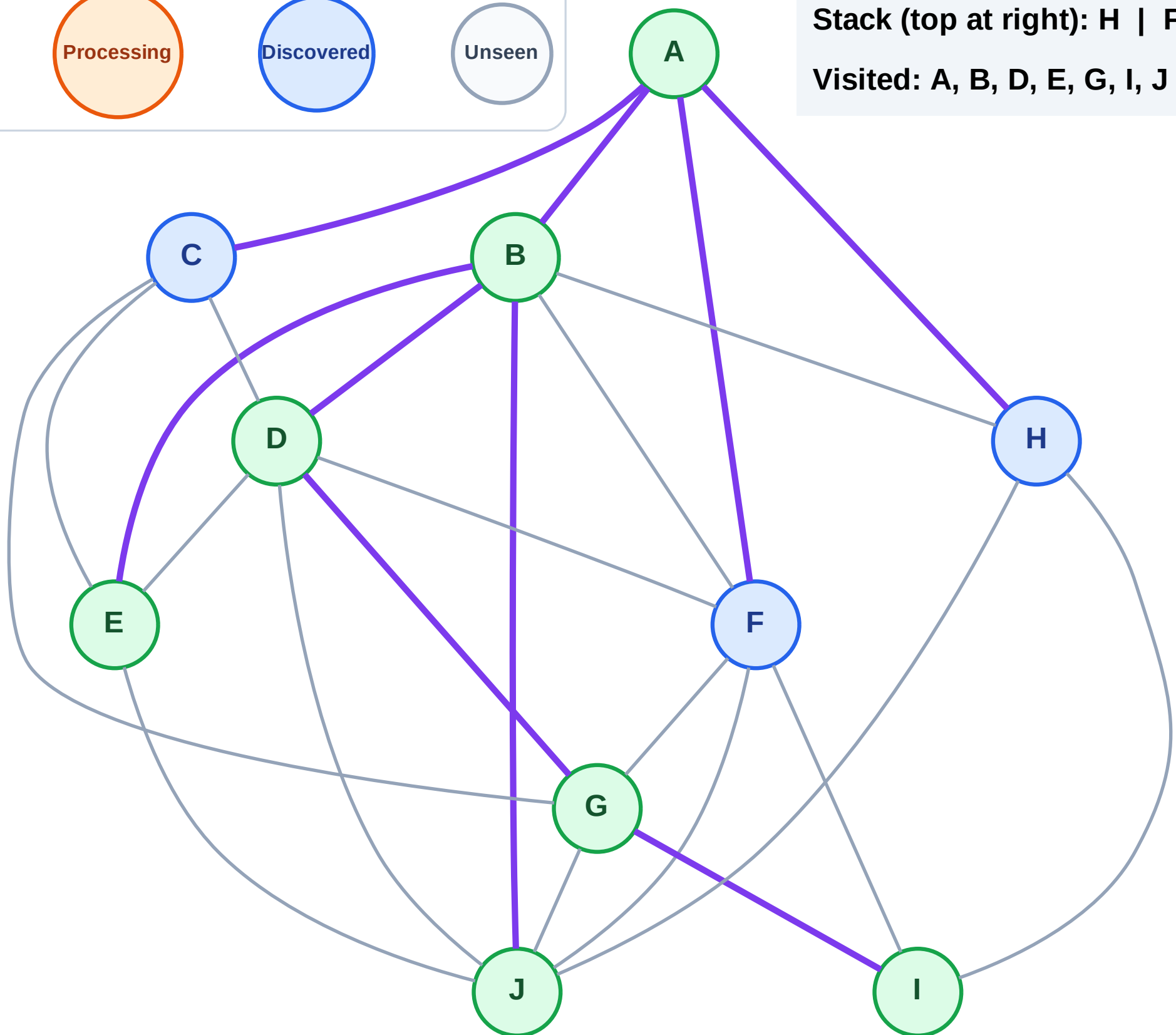
Step 15: No new neighbors from J

Legend



Stack (top at right): H | F | C

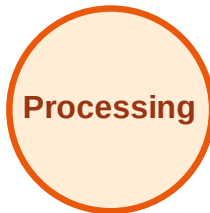
Visited: A, B, D, E, G, I, J



DFS Traversal

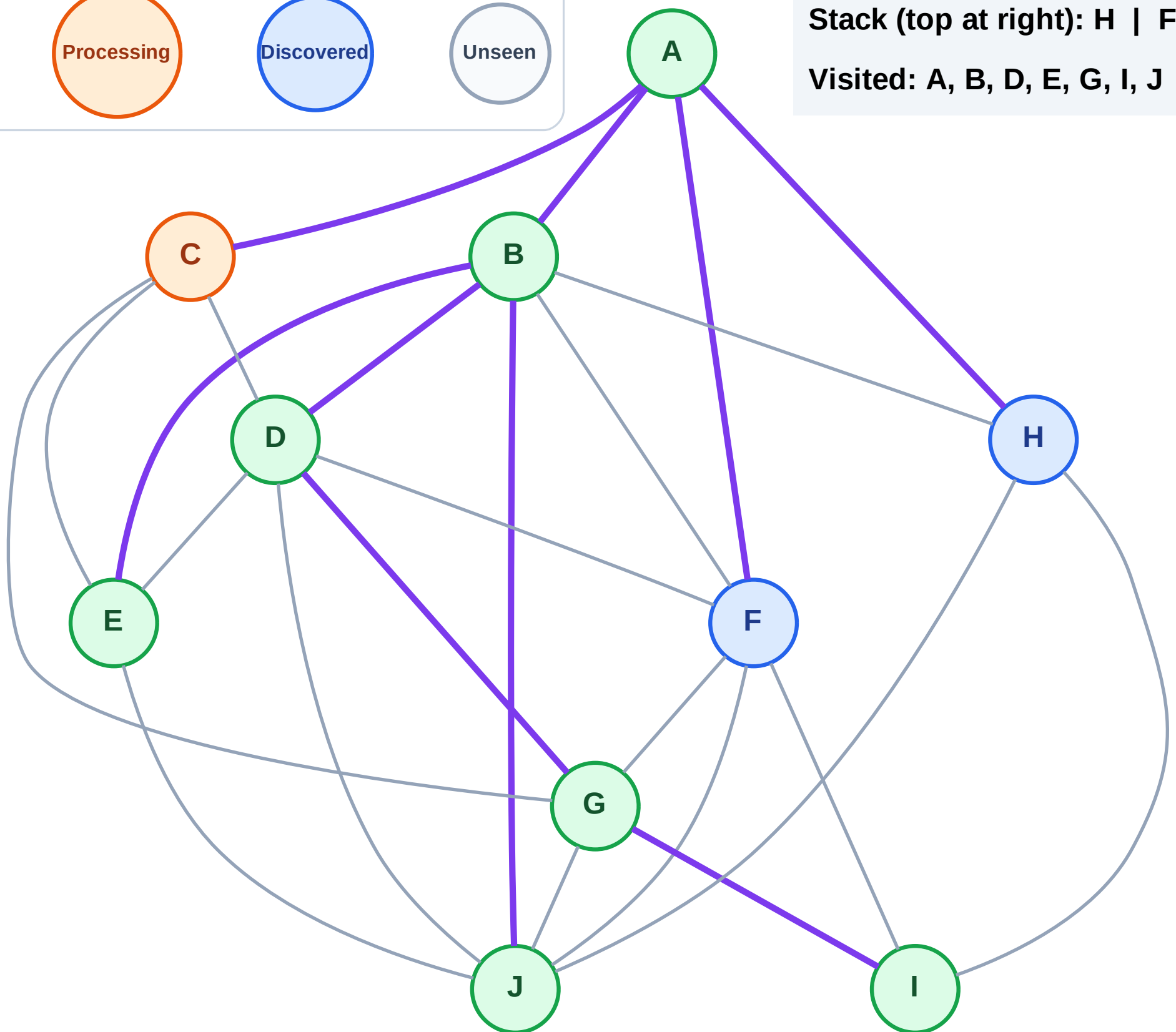
Step 16: Process node C

Legend



Stack (top at right): H | F

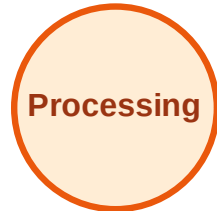
Visited: A, B, D, E, G, I, J



DFS Traversal

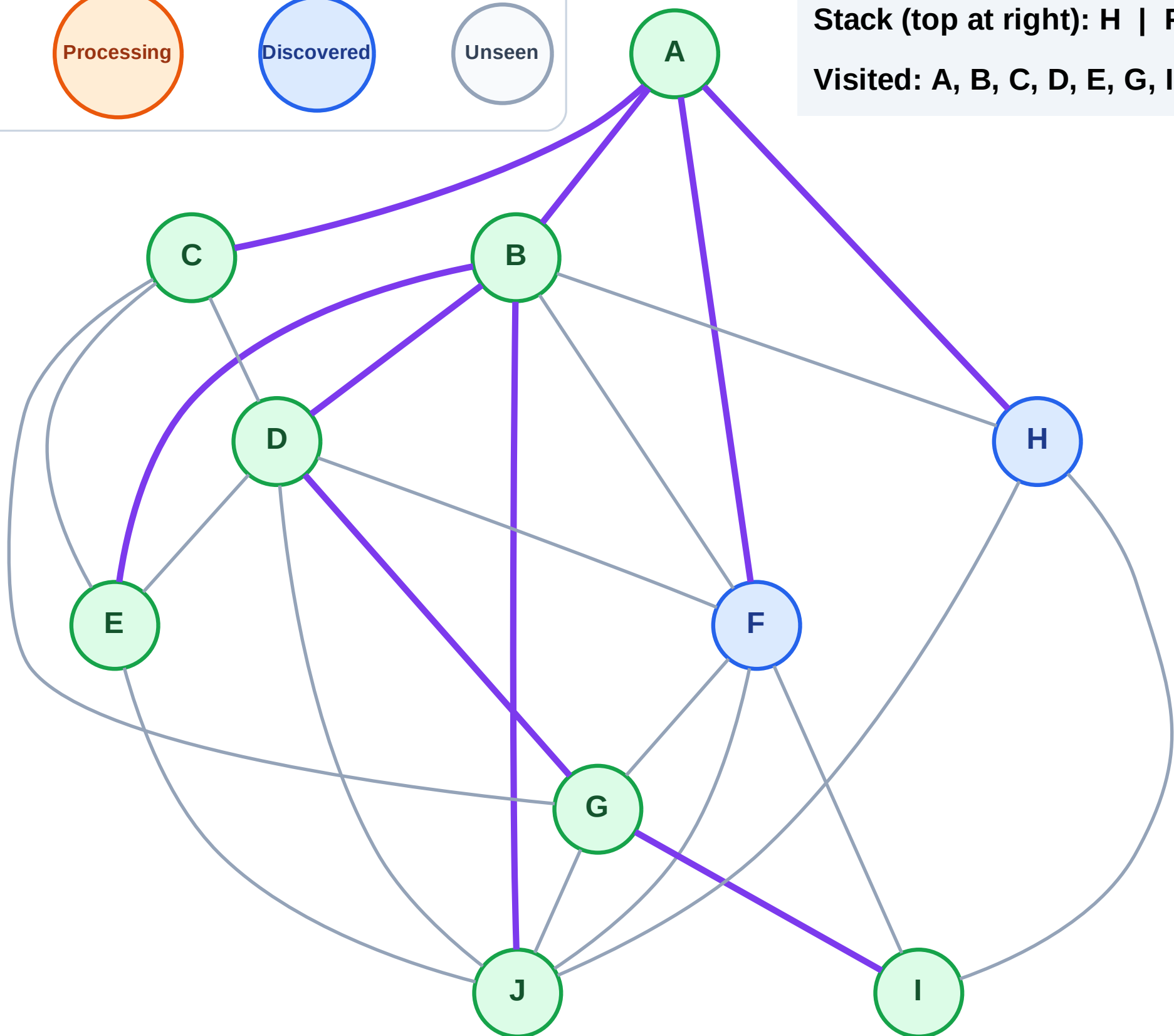
Step 17: No new neighbors from C

Legend



Stack (top at right): H | F

Visited: A, B, C, D, E, G, I, J



DFS Traversal

Step 18: Process node F

Legend

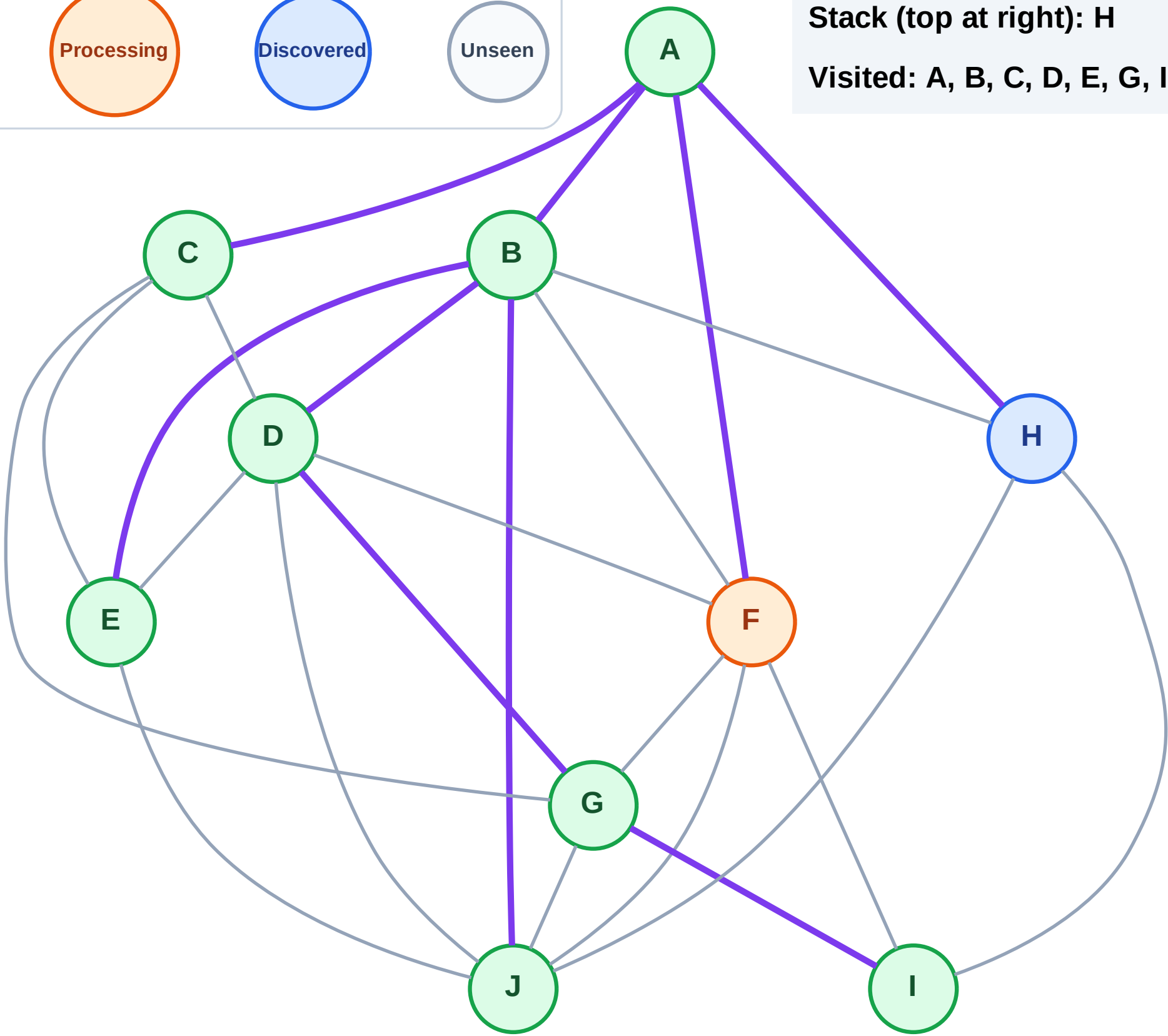
Complete

Processing

Discovered

Unseen

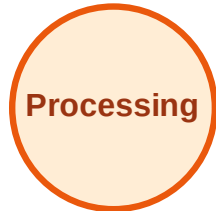
Stack (top at right): H
Visited: A, B, C, D, E, G, I, J



DFS Traversal

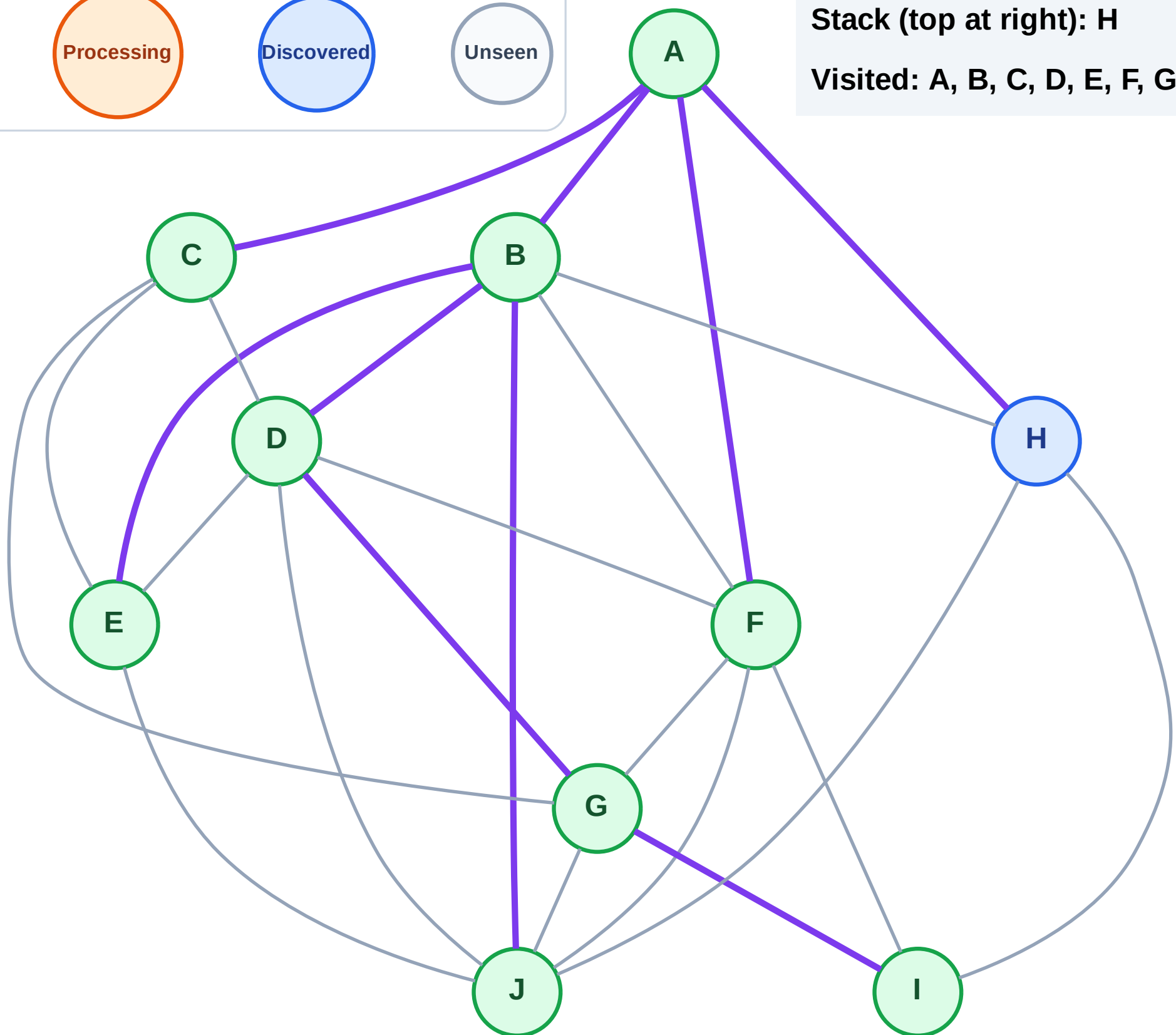
Step 19: No new neighbors from F

Legend



Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J

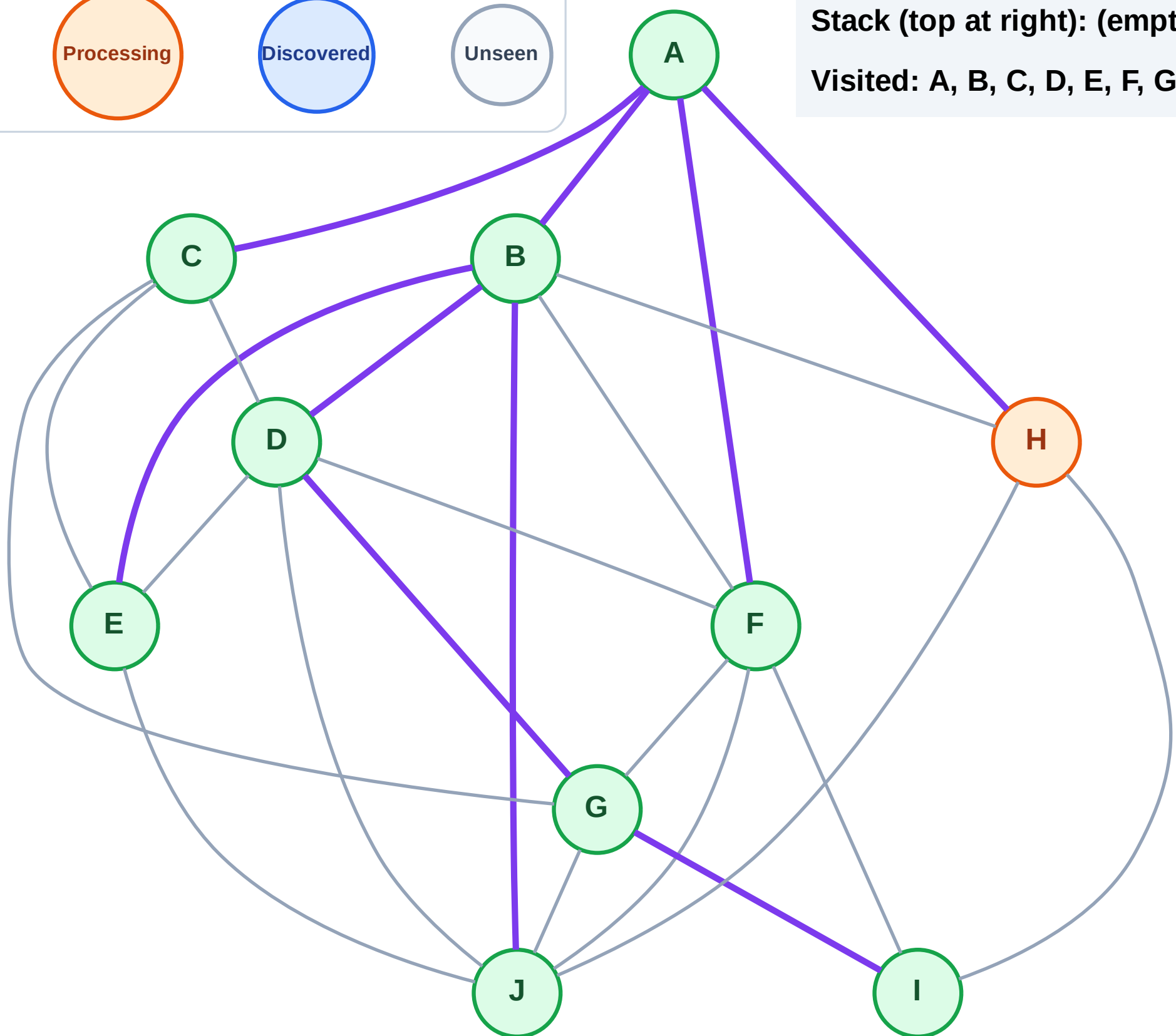


Step 20: Process node H

Legend

Unseen

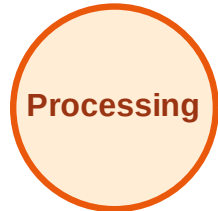
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

